Methodological Framework, Research Progress, and a Usable Interface for PETW in Near-Surface Downburst Wind-Field Simulation with the Cloud-Resolving Idealized Model CM1: A Review

Step 1 CM1 Multi-Model Review Working Group

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Abstract

The intense near-surface outflow of downbursts constitutes the category of convective high-wind hazard that poses the most direct threat to transmission towers, transmission lines, and building structures, yet a long-standing gap persists between the atmospheric-science description of its transient three-dimensional structure and the wind-engineering demand for design gusts. This study takes the CM1 model, operating within a cloud-resolving, nonhydrostatic, compressible moist equation set framework, as its central thread and systematically reviews the formation mechanisms and basic dynamics of downbursts, the historical evolution of simulation method families, the detailed methodological framework and methodological subcategories of CM1, research progress from direct CM1 studies, the reproducibility of initial and numerical settings, the methods, results, and limitations of the Moore (2024) ninety-seven-member idealized ensemble, the validation tiers reached by comparisons of simulations with observations and physical experiments, every mismatch in near-surface wind-field diagnostics, and an operable but unapproved PETW interface pathway from CM1 output to engineering gusts. Throughout, quantitative results are presented in source-bounded language, and all numerical values retain their configuration, diagnostic definition, sample population, and limitations. This study does not claim that PETW has reproduced or validated any downburst case, does not provide a universal accuracy percentage, does not equate wind at 50 m or the lowest model level with a 10 m 3 s engineering gust or `Vhat_3s`, the PETW interface remains `not_approved`, and the final status remains `pending_peter_scientific_review`.

1 Introduction and Literature Review

Convectively driven high-wind events are among the primary meteorological causes of transmission-tower collapse, transmission-line galloping and breakage, and low-rise building damage worldwide, and downbursts are particularly difficult to defend against because they occur abruptly outside the path of the parent storm, produce steep near-surface wind-speed increases, and have small spatial scales (Wakimoto 2001; Qu and Ji 2013). The response of engineering structures to wind loading depends on the gust amplitude, duration, directional change, and spatial correlation

at heights of tens of meters above the surface, and these quantities are precisely the aspects of downbursts that are most difficult for operational observation networks to capture reliably. Wind-engineering practice has long used the stationary strong-wind model of the atmospheric boundary layer as the design baseline, but downburst outflow is an inherently nonstationary, shallow, transient radial flow whose peak wind commonly occurs within the lowest one hundred meters, whose burst swath (the spatial scale of the localized strongest-wind band) is on the order of one hundred meters, and which has been repeatedly recorded to produce near-surface gusts exceeding the National Weather Service (NWS) severe-wind threshold of 25.7 m/s (approximately 26 m/s), with rare cases reaching as high as 67 m/s as cited from Fujita (1985) (Moore 2024). This combination of scale and amplitude makes the conventional design framework based on stationary boundary-layer wind spectra difficult to extrapolate directly to downburst conditions.

Downbursts are classified by outflow diameter into microbursts (4 km and below) and macrobursts (greater than 4 km) and are further distinguished by whether precipitation reaches the surface into dry and wet pathways (Moore 2024). This classification originates from the Fujita system and has been extended in subsequent observational and environmental studies; dry downbursts are more common in deep well-mixed layers over plateaus and semi-arid regions, whereas wet downbursts are more common in southeastern-type environments with higher moisture content and lower cloud bases (Romanic et al. 2022; Hadavi and Romanic 2024). The atmospheric-science community has developed a relatively mature observational and numerical characterization of downburst mechanisms, life cycles, and outflow structure, but these descriptions are based primarily on radar reflectivity, Doppler radial velocity, automated surface station elements, and idealized simulation diagnostics, and a verified, auditable conversion chain to the design gusts required by wind engineering at 10 m height, 3 s averaging period, specified roughness, and return-period conditions is lacking. This gap is particularly acute in transmission-line engineering: transmission towers extend linearly along corridors, their exposure to localized strong wind has spatial cumulative properties, and the narrow-band outflow of a downburst may cut across a corridor to produce loading at one or a few towers far exceeding the design assumption; the formation and dissipation processes of downbursts thus become a dedicated subject of study for the hazards they pose to transmission towers (Qu and Ji 2013). Converting the transient outflow wind fields produced by atmospheric models into near-surface gusts usable for structural assessment requires item-by-item alignment in height, averaging period, roughness, and spatial sampling, and these alignment steps are the scientific core to which this study repeatedly returns.

From a climatological perspective, weakly forced thunderstorms that produce wet downbursts occur frequently during the warm season, yet the proportion that produces severe near-surface wind, the spatial extent and duration of that severe wind, and whether modern observation networks can reliably capture these winds have long lacked systematic quantitative characterization (Moore 2024). The limitation of observation-network spatial scale is clearly reflected in the mean station spacings of several United States surface networks: ASOS/AWOS approximately 44 km, Oklahoma Mesonet approximately 30 km, New York State Mesonet approximately 27 km, MesoWest approximately 15 km, ARS Micronet approximately 5 km, and Oklahoma City Micronet approximately 3 km; the literature recommends a downburst observing network spacing

of 10 to 25 km (Moore 2024). Even at spacings as dense as a few kilometers, relative to the burst swath on the order of one hundred meters (denoting the localized strongest-wind band scale) and the severe-wind footprint on the order of ten to one hundred square kilometers (denoting the order of Moore 50 m swath threshold coverage area), observation networks still systematically undersample, and this constitutes an obstacle both to understanding downburst climatological characteristics and to providing event-level near-surface wind constraints for engineering.

The spatial scale of observational capability is one of the core aspects of this gap. Early downburst field observation networks were mostly meso-γ scale (2 to 200 km) with typical station spacings of 2 to 4 km, sufficient to characterize the overall outflow morphology but potentially too coarse to capture the misoscale (approximately 0.1 to 10 km) burst swath; Doppler radar alleviates the sparseness of surface station networks but, limited by the lowest elevation angle and Earth curvature, often cannot sample the height layer where near-surface peak wind resides (Moore 2024). As the station spacings listed above indicate, even a relatively dense mesoscale network has spacings that are large relative to the severe-wind swath of a downburst, and this mismatch between station spacing and burst-swath scale is the direct source of operational observational undersampling (Moore 2024). Operational observations therefore systematically underestimate the spatial frequency of severe downburst winds and provide limited event-level near-surface wind constraints for engineering. A review of numerical-model limitations and future evolution further notes that mesoscale and microscale models each have their inherent constraints in effective resolution, near-surface closure, and terrain treatment (Hawbecker et al. 2021).

It is precisely these limitations of observational capability that make numerical simulation a necessary means for characterizing the spatiotemporal distribution of downburst near-surface winds. Idealized high-resolution simulations can repeatedly generate downbursts in a controlled environment, outputting continuous three-dimensional wind, thermodynamic, and hydrometeor fields, thereby supplementing the sparse sampling of observations in a statistical sense. This study centers on the cloud-resolving idealized model CM1 and proceeds to discuss the formation mechanisms and basic dynamics of downbursts, the lineage and historical evolution of simulation method families, why CM1 is suited to this problem and its detailed methodological framework, the methodological subcategories and research progress of CM1 studies, the reproducibility of initial conditions and numerical settings, the Moore (2024) ensemble core case study, the validation tiers achieved by comparisons of simulations with observations and physical experiments, the specific problems of near-surface wind-field diagnostics, and how CM1 results can be used by PETW while maintaining unapproved status. This study adopts a method-first, results-specific, limitations-explicit orientation in its writing, all quantitative statements retain their source boundaries, and the engineering conversion chain remains `not_approved`.

2 Formation Mechanisms, Characteristics, and Basic Dynamics of Downbursts

A downburst is a strong downdraft that bursts from a convective parent and, upon reaching the surface, rapidly spreads radially along the ground; the core of its formation is the generation and maintenance within the convective storm of a downdraft possessing significant negative buoyancy and the downward transport of its momentum to the surface (Moore 2024; Wakimoto 2001). The sources of negative buoyancy differ in weight between the wet and dry pathways. In wet downbursts, the primary sources of negative buoyancy are evaporative cooling and latent heat absorption through melting of precipitation particles, while precipitation drag (the gravitational drag of hydrometeors on the air) provides additional downward force to the downdraft; in dry downbursts, within deep dry-adiabatic mixed layers, strong evaporative cooling of raindrops during their descent is dominant, and precipitation largely evaporates before reaching the surface (Romanic et al. 2022; Hadavi and Romanic 2024). The environmental thermodynamic structure has discriminating significance for the two pathways: dry downbursts correspond to a near-dry-adiabatic lapse rate of approximately 9.0 to 10.0°C/km from the surface to 3 km and a low-level dry "inverted-V" profile (as cited from Wakimoto), whereas wet downbursts correspond to smaller lapse rates, higher low-level relative humidity, and a shallower sub-cloud layer (Moore 2024). Entrainment (the ingestion of ambient dry air into the downdraft) amplifies negative buoyancy in both pathways by enhancing evaporation, but its quantitative contribution cannot be itemized within the frozen corpus, and it is stated here only as a mechanistic description.

From an energetic perspective, the downdraft available potential energy is commonly measured by the downdraft convective available potential energy (DCAPE, the negative buoyancy work obtainable by a parcel descending moist-adiabatically from a given height to the surface), and the actual achievable downdraft intensity is jointly determined by the environmental lapse rate, low-level moisture deficit, and mid-level dry-air entrainment, while DCAPE and the actual outflow peak do not have a simple one-to-one correspondence, because entrainment, precipitation drag, and compensating heating during descent all modify the final surface wind (Romanic et al. 2022; Hadavi and Romanic 2024). The relative weights of precipitation drag and evaporative cooling in the wet pathway, and the evaporation-dominated negative buoyancy generation in the dry pathway, vary considerably across environments, and their quantitative decomposition is not available within the frozen corpus of this study; they are treated here as mechanistic statements without giving quantitative shares for each component.

The negatively buoyant air parcel accelerates downward under gravity, forming a concentrated downdraft core. As the downdraft approaches the surface, its vertical momentum can no longer continue downward, and the flow turns horizontally in the near-surface layer, forming a radially outward flow along the ground. The leading edge of the outflow exhibits density-current (gravity-current) characteristics because the cold air is denser than the ambient air, with a lifting head and internal return flow constituting the classic density-current head structure. Behind the head, the cold pool continues to expand outward and pushes the gust front forward; as the gust front passes a fixed point, the wind speed at that point rapidly increases and reaches its peak, then decays in

an oscillatory manner of decreasing amplitude (Moore 2024; Wakimoto 1982). Between the upper outflow and the cold-pool edge, vertical wind shear commonly rolls up horizontal vortex tubes, forming a ring vortex or local rotor structure, and this structure directly influences the location and intensity of the near-surface peak wind (Hjelmfelt 1988). The ring vortex propagates outward with the outflow and gradually lifts, stretches, and dissipates; its localized acceleration near the gust-front head can cause the near-surface wind to reach its strongest value at the instant the head passes, and this head-enhancement effect has been clearly documented in Doppler observations of the gust-front life cycle (Wakimoto 1982). The propagation speed of the density current is controlled by cold-pool depth and temperature deficit; the deeper the cold pool and the greater the temperature deficit, the faster the gust front advances and the stronger the near-surface wind, and this relationship is the mechanical foundation for understanding the spatiotemporal evolution of outflow.

Several characteristics of the downburst near-surface wind field are particularly critical for engineering applications. First is the shallow low-level wind maximum: the strongest horizontal wind typically occurs within the lowest one hundred meters, observations document the peak mostly at 50 to 200 m AGL, in idealized simulations the strongest wind within the gust front can be located at approximately 50 m AGL, and above 250 m the wind speed drops rapidly below 10 m/s, indicating that the outflow wind is vertically shallow (Moore 2024). Second is asymmetry: in the presence of ambient wind or advection, the strongest wind is displaced to the downwind side, the outflow exhibits an asymmetric distribution, and a classic rotor is visible on the upwind edge of the cold pool (Moore 2024). Third is transience and nonstationarity: the time scale from onset to decay of the outflow is on the order of minutes, the wind-speed time series is markedly nonstationary, and it cannot be described using stationary strong-wind statistical assumptions. These characteristics have been repeatedly confirmed in observations and multiple numerical studies and represent relatively established mechanistic understanding; however, the specific peak amplitude, peak height, and degree of asymmetry depend strongly on environment and configuration and represent source-specific findings that should not be treated as mechanistically universal constants.

The scale distinction between microbursts and macrobursts is not merely a geometric classification but also corresponds to differences in outflow structure and life cycle: microbursts are more concentrated and transient, with an outflow diameter of 4 km or less and duration commonly on the order of minutes; macrobursts have broader and longer-lasting outflow, with diameters greater than 4 km. This scale spectrum means that the spatial correlation structure of near-surface wind varies significantly with downburst type, and engineering exposure assessments for corridor-type and point-type structures therefore cannot adopt a uniform spatial-averaging assumption. Downbursts can occur in weakly forced isolated single-cell convection (whose experimental design most closely approximates this situation) or can be embedded within organized convective systems such as squall lines and bow echoes, in which cold-pool outflow and system-scale dynamics superpose with the downburst, producing more complex near-surface strong-wind distributions; the system-scale organization and cold-pool outflow dynamics of long-lived squall lines and severe bow echoes provide the background for this superposition (Rotunno et al. 1988; Weisman 1993).

Within such systems, the generation of low-level mesovortices and the descent of the rear-inflow jet can superpose system-scale strong winds with local downbursts, making the causal attribution of near-surface strong wind complex, and the strong winds produced by these low-level mesovortex and rear-inflow mechanisms should be distinguished from the buoyancy-driven outflow of isolated downbursts (Trapp and Weisman 2003; Atkins and Laurent 2009; Wakimoto et al. 2006). The organizational modes of mesoscale convective systems themselves also influence the occurrence frequency and distribution of downbursts (Zheng et al. 2013).

Regarding environmental parameters, CAPE and DCAPE characterize the convective available energy and the potential negative buoyancy of the downdraft, and deep-layer and low-level shear influence convective organization and outflow symmetry, but in weakly forced wet downburst samples, the discriminating power of MLCAPE and deep-layer shear (EBWD) between severe and sub-severe events is weak, with highly overlapping interquartile ranges (Moore 2024; Romanic et al. 2022). The total totals index (TT), temperature-dewpoint-related quantities (TED), and mixed-layer lifted condensation level (MLLCL) show some discriminating ability, and surface-based and most-unstable convective inhibition (SBCIN, MUCIN) also exhibit some discrimination, but the median convective inhibition for both event categories is zero, indicating that convective inhibition alone is insufficient to discriminate whether severe wind is produced (Moore 2024). The overlapping interquartile ranges of these parameters indicate that setting a single environmental threshold has limited utility for operationally forecasting severe downbursts; environmental conditions are better suited as necessary but not sufficient criteria for whether convection may be damaging. This discriminating-power limitation is important background for the validation and PETW discussions in subsequent sections, and it also explains why characterizing near-surface wind distributions with an ensemble approach covering diverse environments is more reliable than extrapolating from a single environmental parameter.

3 Lineage and Historical Evolution of Downburst Simulation Methods

Downburst simulation methods can be classified by their treatment of physical processes, including their resolution, prescription, omission, and validation, into several families, with CM1 situated at the cloud-resolving idealized end. At the simplest end are analytical and semi-empirical models, which prescribe the outflow structure directly using a given velocity field or density-perturbation function, are extremely inexpensive computationally, and are convenient for engineering parameterization and preliminary load estimation, but they do not solve for the convective generation process and cannot self-consistently produce the formation and evolution of the downdraft; their radial profile, vertical profile, and temporal evolution are all input as a priori assumptions rather than model results. These models are widely used in engineering to generate design wind fields, but their ability to represent asymmetry, collision, and transient structures is limited, and their parameters must be calibrated by higher-fidelity simulations or observations.

The physical-experiment approach is dominated by the impinging-jet concept, which repro-

duces the radial spreading and leading-edge vortex ring of outflow through a circular jet impinging on a flat plate. Impinging-jet experimental and numerical comparison studies have characterized the radial velocity distribution and vortex-ring generation after jet impact on a plate (Sengupta and Sarkar 2008); the proposed large-scale modeling of transient downburst outflow features has explored how to reproduce the nonstationary outflow of downbursts in experiments (Lin et al. 2007); the WindEEE Dome downburst-type impinging-jet experiment has provided a high-resolution near-surface measurement dataset (Canepa et al. 2022); and a comparative study of impinging-jet and cooling-source microburst modeling has systematically compared the similarities and differences between the outflow produced by the two methods (Zhang et al. 2013). The impinging-jet approach can study outflow kinematics, vortex-ring evolution, and surface-roughness effects in a controlled manner and can provide dense spatial and temporal sampling, but it replaces the buoyancy-driven downdraft with a mechanical jet boundary condition and lacks the real thermodynamic generation process, so its similarity to real downbursts is not automatic and must be verified on a non-dimensional parameter by non-dimensional parameter basis.

Among numerical methods, the cooling-source approach creates negative buoyancy in a specified region within a computational fluid dynamics framework using an externally imposed space-time cooling function, thereby triggering a downdraft and outflow without explicitly solving cloud microphysics; it is computationally relatively economical and is suited to parameter sweeps and scaling-law studies (Oreskovic et al. 2018b; Kondo 2021; Oreskovic 2016; Vermeire et al. 2011a,b). This class of methods has been applied under various turbulence treatments including CFD, RANS, URANS, and LES; its advantage is strong controllability, and its limitation is that it replaces real precipitation evaporation, melting, and drag with prescribed cooling and therefore cannot self-consistently reproduce the complete thermodynamic and hydrometeor structure of wet downbursts. A study aimed at improving cooling-source modeling for wind engineering explicitly demonstrated that the near-surface flow topology of the CM1 cooling source and impinging jet in the tested configuration exhibits significant differences, with a non-dimensional vorticity-mapping deviation of at least 55%, and that the aspect ratio and height of the ring vortex are strongly correlated with surface roughness length whereas the impinging jet is essentially roughness-insensitive, indicating that the two simplified methods cannot substitute for or directly calibrate each other (Vermeire et al. 2011a).

On the atmospheric-model line, predecessor models provided mechanistic and methodological groundwork for CM1. Proctor's first numerical simulation system of isolated microbursts systematically characterized the dynamics and structure of the downdraft, establishing the basic framework of explicit-microphysics-driven idealized microburst simulation (Proctor 1988); the companion sensitivity experiments examined the influence of environmental and microphysics parameters on microburst intensity and structure, revealing the controlling roles of lapse rate, humidity, and precipitation particle spectra on outflow (Proctor 1989). An early numerical study of severe thunderstorm gust fronts revealed the dynamics of density-current head lifting, internal return flow, and gust-front leading-edge processes, laying the mechanistic foundation for subsequent simulation of gust fronts and cold-pool outflow (Mitchell and Hovermale 1977). These predecessor models established the feasibility of simulating downbursts with explicit microphysics

and nonhydrostatic dynamical cores and also established lapse rate, humidity, and shear as key controlling parameters, but their specific grid, closure, and calibration parameters should not be treated as directly inherited by CM1; the specific settings of models such as TASS and ARPS are not supported in the frozen corpus of this study and are marked as NR, cited only in the sense of lineage connection for their mechanistic contributions. The mesoscale event-model line is represented by WRF, which performs regional simulations targeting real damaging events for mechanism diagnosis and event reconstruction. Specific work along this line includes: detecting microbursts with WRF and evaluating their effective resolution and forecasting indices (Bolgiani et al. 2020b), simulating atmospheric microbursts at high spatiotemporal resolution with a mesoscale model (Bolgiani et al. 2020a), analyzing the simulation of a downburst event causing a transmission-tower collapse (Li et al. 2023), simulating the formation and evolution mechanisms of a destructive-wind thunderstorm (Liu et al. 2023), implementing a gust-front head-collapse scheme in WRF to improve near-surface strong-wind representation (Lompar et al. 2018), and investigating weather conditions during a bridge collapse (Burlando et al. 2020). The advantage of the WRF event line is its link to real observations and geographical context, enabling reconstruction or diagnosis of the synoptic-scale background, triggering, and evolution of events; its limitation is that its grid is typically coarse (on the order of 4 km), making it difficult to correctly resolve near-surface peak winds at the burst-swath scale, and its near-surface strong wind often requires dedicated downscaling or scheme improvements to approach observational magnitudes. This limitation is precisely where the idealized high-resolution CM1 line is complementary: CM1 trades correspondence with a specific real event for the ability to resolve near-surface peak wind and fine outflow structure.

The cloud-resolving idealized model CM1 is situated at the high-resolution end of this lineage. It solves convective-scale dynamics using the nonhydrostatic, compressible moist equation set, combined with explicit microphysics or a prescribed cooling source, and generates downbursts in an idealized homogeneous environment triggered by a warm bubble or cooling function. Two benchmark works for CM1 respectively established its conservation properties as a moist nonhydrostatic model (the fully retained equation set in a two-dimensional rising thermal bubble benchmark produces mass and energy relative errors on the order of approximately $10^{-4}\%$, and equation-set approximations significantly alter thermodynamic evolution) and the resolution requirements for deep moist convection (in idealized studies, grids of 1 km, 500 m, 250 m, and 125 m did not converge, an inertial subrange begins to appear only at approximately 250 m and finer, and O(100 m) is posed as a hypothetical suggestion rather than a convergence guarantee) (Bryan and Fritsch 2002; Bryan et al. 2003). Figure 1 presents the relative positioning of the above method families in terms of time and near-surface wind resolving power as an original synthesized schematic. The suitability of CM1 derives from its ability to explicitly solve downdraft generation, outflow evolution, and near-surface structure, but the conversion between its near-surface diagnostics and engineering gusts still requires dedicated treatment, a point that will be elaborated in subsequent sections.



Figure 1: Schematic of the evolution and relative positioning of downburst simulation method families. This figure is an original schematic newly synthesized from the literature cited in this study and reproduces no source figure. The concepts it synthesizes are drawn from analytical and semi-empirical models, impinging-jet physical experiments (Sengupta and Sarkar 2008; Canepa et al. 2022; Zhang et al. 2013; Lin et al. 2007), cooling-source methods (Oreskovic et al. 2018b; Kondo 2021; Oreskovic 2016; Vermeire et al. 2011a), predecessor atmospheric models (Proctor 1988, 1989; Mitchell and Hovermale 1977), WRF event models (Bolgiani et al. 2020b,a; Li et al. 2023; Liu et al. 2023; Lompar et al. 2018; Burlando et al. 2020), and the CM1 cloud-resolving idealized model (Bryan and Fritsch 2002; Bryan et al. 2003; Moore 2024). The horizontal axis indicates the approximate order of appearance of each method, the vertical axis indicates resolving power for transient near-surface outflow, and the relative positions are qualitative schematic rather than a quantitative scale.

4 Why CM1 Is Suited to Downburst Simulation and Its Detailed Methodological Framework

The fundamental reason CM1 is suited to the downburst problem is that this problem requires the simultaneous resolution of convective-scale downdraft generation, density-current-type near-surface outflow evolution, and transient near-surface wind fields within the lowest hundred meters, and all three fall within the scale regime in which the cloud-resolving, nonhydrostatic, compressible moist equation set framework excels. This section proceeds field by field through the methodological framework; wherever the frozen corpus does not report a field, its unreported status is stated in prose and marked as NR, without back-inference from model defaults, and equations are used only where an authoritative source supports their form, with symbols defined at first occurrence.

The reason CM1 is suited to the downburst problem is that the problem physically demands the simultaneous satisfaction of three conditions: the self-consistent generation and maintenance of a downdraft possessing significant negative buoyancy, the resolution of the shallow and transient radial density-current outflow after surface impact, and the diagnosis of peak-wind structure within the lowest hundred meters in sufficiently high-resolution configurations, with some studies resolving near-surface gradients in the lowest tens of meters. These three conditions respectively point toward convective-scale moist thermodynamics, gravity-current dynamics, and high-resolution near-surface turbulence, and the cloud-resolving, nonhydrostatic, compressible moist equation set framework simultaneously solves these processes in precisely this scale regime, without requiring a hydrostatic approximation for vertical acceleration or parameterization for convection. Compared with coarser mesoscale models based on the hydrostatic approximation, CM1 retains the full nonhydrostatic terms of the vertical momentum equation, thereby enabling representation of the strong vertical acceleration within the downdraft core; compared with impinging-jet or pure cooling-source CFD driven by prescribed boundary conditions, CM1 in its wet full-storm configuration can self-consistently produce the thermodynamic generation of the downdraft.

Regarding the framework category, CM1 solves the cloud-resolving, nonhydrostatic, compressible moist atmospheric equation set. Its conservation properties have been examined in a two-dimensional rising thermal bubble benchmark: under the fully retained equation set, the relative errors of mass and energy are on the order of approximately $10^{-4}\%$, and different equation-set approximations (such as varying degrees of acoustic or mass-conservation simplification) significantly alter thermodynamic evolution, indicating that the choice of equation set is itself a physical decision rather than a purely numerical detail (Bryan and Fritsch 2002). This benchmark is configuration-specific conservation evidence for a two-dimensional rising thermal bubble and cannot be taken as a convergence proof for three-dimensional downburst problems. Moore (2024) used CM1 Release 20 version 3 as the dynamical core for the ensemble (Moore 2024), and the studies reviewed overall span CM1 r18, cm1r19.10, r20.3, and v13, with version differences themselves indicating that numerical details cannot be assumed consistent across studies. Regarding base-state and perturbation treatment, the framework adopts a base-state-plus-perturbation variable

decomposition, decomposing thermodynamic and pressure variables into a horizontally homogeneous base state and three-dimensional perturbations, with the dynamical equations solved for the perturbation quantities, but the reviewed downburst studies do not provide in their text the exact base-state definition (isentropic base state, hydrostatic base-state profile, or specific decomposition form of perturbation variables), so this field is NR; the decomposition concept is stated here in prose without giving unsupported equations.

Regarding prognostic variables, CM1 predicts the three velocity components u, v, w (east-west, north-south, and vertical wind speeds, respectively), thermodynamic variables, and hydrometeors. In wet full-storm simulations using explicit microphysics, the Morrison double-moment scheme predicts the mixing ratio and number concentration of rain, snow, graupel, and hail; double-moment means that both the mass mixing ratio and the number concentration are predicted simultaneously for each hydrometeor category, thereby better representing the particle size spectrum and fall speed, which in turn affects the spatial distribution of precipitation drag and evaporative cooling (Moore 2024; Orf et al. 2014). This point is particularly critical for downbursts because the generation of negative buoyancy directly depends on the evaporation and melting of precipitation particles during descent, which in turn depends on the particle size spectrum. In cooling-source dry idealized studies, an externally imposed cooling function replaces explicit microphysics, hydrometeor phase changes are not predicted, and the cooling function creates negative buoyancy with a specified spatiotemporal shape and peak cooling rate, thereby compressing the complex microphysics processes into a small number of controllable parameters (Oreskovic et al. 2018b; Kondo 2021; Oreskovic 2016; Vermeire et al. 2011b,a). The choice between these two prognostic-variable schemes directly determines the scientific questions the simulation can address: explicit microphysics is suited to studying the self-consistent evolution of real wet downbursts, and the cooling source is suited to studying the response of outflow to a small number of controlling parameters and scaling laws. Regarding pressure and buoyancy coupling, evaporative cooling and hydrometeor drag jointly drive negative buoyancy downdraft, buoyancy perturbations couple through the pressure and velocity fields, and pressure perturbations play an important role in accelerating the near-surface wind at the point of downdraft surface impact and outflow turning (Moore 2024); however, the exact pressure-solver form (split-explicit acoustic-step integration or the specific form of a semi-implicit pressure equation) is NR in the text of the reviewed studies, and the coupling relationship between pressure and buoyancy is therefore stated here in mechanistic prose without writing a pressure equation.

Regarding grid and staggering, the reviewed studies universally adopt a staggered velocity grid (Moore citing Armfield 1991) and a Cartesian grid, with horizontal and vertical spacings varying by study (Moore 2024). Moore's ensemble employs an 80 km $\times$ 80 km horizontal domain, 0.1 km horizontal spacing, 15 km model depth, 84 vertical levels, constant 0.1 km vertical spacing below 2 km, stretching from 2 to 9 km, and constant 0.25 km from 9 to 15 km, placing the lowest model level at approximately 50 m AGL (Moore 2024). Cooling-source parametric studies employ finer near-surface grids, such as a constant 10 m horizontal grid combined with a vertical grid stretched from 1 m near the surface (CM1 r18) (Oreskovic et al. 2018b), or a 1 to 50 m stretched vertical grid (Oreskovic 2016), and another cooling-source study uses a 0.05 s time step and 10 m grid

(cm1r19.10) (Kondo 2021). High-resolution three-dimensional wet simulations employ grids of approximately 20 m horizontal and approximately 5 m vertical near the surface at the downburst center (CM1 v13), with the circumferential analysis study employing a 92 km × 92 km × 14 km domain (Orf et al. 2012, 2014; Oreskovic et al. 2018a). Regarding spatial numerics, Moore's ensemble employs sixth-order horizontal, fifth-order vertical advection, and sixth-order diffusion (Moore 2024); the cooling-source LES study employs sixth-order explicit horizontal advection (Oreskovic et al. 2018b); and one study reports that highly symmetric radial outflow under odd-order advection triggers a non-physical clefting artifact on the Cartesian grid, which is eliminated by switching to a sixth-order explicit horizontal scheme combined with tuned diffusion; this is a configuration-specific recommendation, not a universal requirement (Oreskovic 2016).

Regarding time integration and acoustic treatment, Moore reports a time step of 1 s and data output every 60 s (Moore 2024), and a cooling-source study reports a time step of 0.05 s (Kondo 2021), but the time-integration scheme (such as third-order Runge-Kutta) and acoustic or compressible-term treatment (split-explicit acoustic step or Klemp-Wilhelmson-type scheme) are NR in the text of the reviewed studies; they are not back-inferred from CM1 defaults, and only the reported time steps are stated here without giving unreported integration schemes. Regarding diffusion, beyond the sixth-order diffusion and tuned explicit diffusion mentioned above, the exact diffusion coefficients are NR. Regarding subgrid turbulence, the reviewed studies employ Deardorff TKE closure (Moore's Table 2 citing Deardorff 1980; cooling-source parametric studies also report Deardorff TKE), which can be described as LES or TKE-type closure, but it should not be claimed that this closure is uniquely sufficient for downbursts or has converged; the relevant studies all set boundaries at this point (Moore 2024; Oreskovic 2016; Bryan et al. 2003). Regarding microphysics, wet simulations employ the Morrison double-moment or Thompson scheme (Moore 2024; Orf et al. 2014; Schumacher et al. 2023), and dry idealized studies use the cooling source as a substitute (Oreskovic et al. 2018b; Kondo 2021; Oreskovic 2016; Vermeire et al. 2011b,a).

Regarding the surface layer and boundaries, Moore employs the MM5 similarity-theory surface-layer scheme, a semislip lower boundary, open radiative lateral boundaries, a free slip upper boundary, the radiation scheme set to None, and the land-use type set to low-intensity residential (Moore 2024); the event-paired study employs a free-slip lower boundary and Thompson microphysics (Schumacher et al. 2023); the cooling-source comparison study employs quarter-domain mirror-symmetry boundaries to save computation (Vermeire et al. 2011a); and one parametric study explicitly turns off the air-ground heat flux (setting zero temperature difference) (Oreskovic 2016). Coriolis setting and land-surface model details beyond land-use type are NR in the text. Regarding input soundings, Moore's soundings are derived from SPC sfcOA (40 km, hourly, RUC/RAP-based), with 25 mb vertical resolution covering the surface to 100 mb (Moore 2024); the circumferential analysis and scaling-law studies employ modified dry downburst soundings (citing modified Brown et al. 1982) (Orf et al. 2014; Oreskovic et al. 2018a); and the event study employs an event sounding (Schumacher et al. 2023). Regarding triggering, Moore uses a single positive 2 K warm bubble, with horizontal width 10 km, vertical depth 1.5 km, initial vertical position 0.5 km, located at the domain center and held fixed across all simulations (Moore 2024); cooling-source studies use an externally imposed cooling function for triggering.

The choice of boundary conditions directly influences the downburst near-surface wind. Open radiative lateral boundaries allow perturbations and outflow to radiate outward from the domain without producing non-physical reflections, which is necessary for capturing outflow evolution in a finite domain on the order of 80 km; the free slip upper boundary avoids upper-boundary friction interference; the lower-boundary treatment is the most critical: the semislip lower boundary partially couples momentum to the surface, and Moore employs it together with the MM5 similarity surface layer, whereas the event-paired study employs a free-slip lower boundary and consequently significantly overestimates extreme winds, and the cooling-source comparison study employs quarter-domain mirror symmetry to save computation (Moore 2024; Schumacher et al. 2023; Vermeire et al. 2011a). The difference in lower boundary and surface-layer scheme directly alters near-surface wind attenuation and is one of the primary obstacles to comparability of near-surface peaks across studies, also indicating that in the PETW context the lower-boundary treatment must be regarded as a first-order configuration decision rather than a default option. Regarding land use, Moore employs the low-intensity residential type, and other studies are mostly NR; the radiation scheme is None, reflecting the negligible influence of radiation on downbursts in idealized short-duration simulations; Coriolis setting and land-surface model details are NR in the text and are not back-inferred from defaults.

Regarding output and diagnostics, Moore outputs a 50 m maximum-wind swath updated every 1 s and written every 60 s, instantaneous 10 m diagnostic U/V winds every 60 s, and 2 m temperature and humidity (Moore 2024); the update and sampling definitions of these diagnostics differ from one another and are the core of the near-surface wind discussion in subsequent sections. The construction of each diagnostic quantity determines what question it can answer: the 50 m maximum-wind swath, because it is updated every 1 s, can capture the instantaneous peak as the gust front passes and records it as a spatial envelope; the 10 m diagnostic wind, because it is sampled instantaneously every 60 s, may miss peaks between output steps; and the 2 m temperature and humidity are used to characterize cold-pool intensity and the thermodynamic signature of the outflow. Understanding the physical height, update frequency, and sampling definition of each of these diagnostics is the first line of defense against mistaking model diagnostics for engineering quantities, and is the technical core to which Sections 10 and 11 repeatedly return. Regarding conservation and resolution benchmarks, the two-dimensional rising thermal bubble conservation benchmark provides a configuration-specific error on the order of approximately $10^{-4}\%$ as a reference for evaluating the conservativeness of the numerical implementation of the model (Bryan and Fritsch 2002). The resolution-requirement study for deep moist convection demonstrates that in idealized deep moist convection, solutions on grids of 1 km, 500 m, 250 m, and 125 m did not converge, an inertial subrange begins to appear only at approximately 250 m and finer, and the authors therefore pose O(100 m) as a hypothetical suggestion rather than a convergence guarantee (Bryan et al. 2003). The direct implication for downbursts is that even an O(100 m) horizontal grid may not resolve the full turbulent structure of the outflow, so the peak wind obtained at any given grid spacing should be regarded as a diagnostic at that resolution, not as a converged true value. Moore's horizontal grid sensitivity shows that four same-sounding simulations at 0.05, 0.1, 0.2, and 0.5 km horizontal spacings yield maximum 50 m horizontal wind

that decreases with increasing spacing, declining 2.4%, 8.4%, and 12.8% at 0.1, 0.2, and 0.5 km respectively relative to 0.05 km; the 20 m/s threshold coverage decreases logarithmically with spacing, while the 24 m/s threshold coverage is comparable between 0.05 and 0.1 km (Moore 2024). The vertical grid sensitivity shows that three sets of 84, 140, and 150 levels yield maximum 50 m wind differences all within 1.5 m/s, but the areal coverage of 50 m wind above 24 m/s decreases by approximately 25% at 150 levels and approximately 70% at 140 levels; the author attributes this to differences in initial downdraft intensity and size and cites prior work to indicate that such sensitivity is not unexpected (Moore 2024). A counterintuitive yet important result is that increasing grid spacing, while only slightly lowering peak wind, produces a stronger downburst and a larger outflow radius, suggesting that peak wind may be under-resolved at the standard configuration. These results should be understood as sensitivity within a configuration boundary rather than as universal convergence proof; the specific grid of any study (such as 0.1 km or 20 m) should not be treated as a convergence threshold or optimal grid for PETW. In summary, the CM1 framework provides self-consistent and auditable solving capability for downbursts at the dynamical-core, microphysics, boundary, and diagnostic levels; its reported fields (version, grid, advection order, closure, microphysics, boundaries, triggering, diagnostics) are sufficient to support a method-first review, while its unreported fields (time-integration scheme, acoustic treatment, exact pressure equation, Coriolis and land-surface details, exact diffusion coefficients) have been itemized and marked as NR, constituting the explicit uncertainty boundary of this study at the methodological level.

5 Methodological Subcategories of CM1 Downburst Research

After CM1 was adopted as the core for downburst simulation, the relevant studies can be classified by scientific purpose into several methodological subcategories, each corresponding to specific configuration requirements, typical outputs, strengths, limitations, and PETW relevance; quantitative results should not be migrated across subcategories. The first subcategory is dry or prescribed cooling-source outflow studies, which use an externally imposed cooling function to generate outflow under controlled conditions for studying outflow kinematics, scaling laws, and numerical-scheme effects; typical outputs are near-surface radial velocity fields, vortex diagnostics, and peak wind; its strength is controllability and computational economy; its limitation is the replacement of real microphysics with prescribed cooling; and its PETW relevance is primarily as a mechanistic reference for peak-to-mean relations rather than event-level wind fields (Oreskovic et al. 2018b; Kondo 2021; Oreskovic 2016; Vermeire et al. 2011a,b). The second subcategory is explicit-microphysics full-cloud or full-storm simulations, which initialize with a sounding and warm-bubble trigger and employ a microphysics scheme such as Morrison double-moment to self-consistently generate the complete process from convection to downdraft outflow; typical outputs are reflectivity proxy, hydrometeors, three-dimensional wind fields, and near-surface diagnostics; its strength is physical self-consistency; its limitation is computational expense and sensitivity to sounding and trigger; and its PETW relevance is the highest. This subcategory includes both wet downburst simulations initialized with weakly forced wet environment ensem-

bles (Moore 2024; Orf et al. 2012) and high-resolution microphysics simulations initialized with modified dry downburst soundings (Orf et al. 2014; Oreskovic et al. 2018a); although both belong to the explicit-microphysics full-cloud pathway, their initial environments are of the wet type and the modified dry type respectively, their quantitative results should not be conflated, and they should not be collectively labeled as a single wet pathway.

The third subcategory is scaling and parametric studies, which systematically vary cooling rate, lapse rate, or environmental parameters to examine the non-dimensional scaling relationships of outflow velocity and vortex structure; typical outputs are normalized velocity and vortex diagnostic collapse curves; the strength is revealing controlling parameters; the limitation is that scaling relationships hold only within the tested matrix (Oreskovic et al. 2018b; Kondo 2021; Oreskovic et al. 2018a). The fourth subcategory is high-resolution three-dimensional near-surface and circumferential studies, which use extremely high resolution (approximately 20 m horizontal, 5 m near-surface vertical) to resolve the outflow leading edge, ring vortex, and near-surface peak, and evaluate the applicability of the axisymmetric approximation through circumferential averaging; typical outputs are near-surface peak wind, peak factor, and circumferential structure; the strength is resolving near-surface fine structure; the limitation is that most are single or few simulations (Orf et al. 2012, 2014). The fifth subcategory is interaction, collision, or downburst-line studies, which resolve the interaction of multiple downdrafts and their amplification of peak wind and damaging-wind footprint; typical outputs are non-standard outflow profiles and peak amplification in the collision region; the limitation is that collision-region profiles depart strongly from isolated empirical shapes (Vermeire et al. 2011b).

The sixth subcategory is event-paired or WRF-linked mechanism studies, which combine idealized CM1 experiments with regional simulations of real events to diagnose the contribution of specific mechanisms (such as gravity-wave breaking) to strong near-surface winds; typical outputs are mechanism attribution and parcel analysis; the strength is linkage to real events; the limitation is that settings such as free slip may overestimate extreme winds and that event-level peaks cannot be precisely matched (Schumacher et al. 2023). The seventh subcategory is ensemble and synthetic observation network studies, which use tens to hundreds of idealized members to characterize the spatiotemporal distribution of near-surface wind and construct synthetic observation networks to evaluate sampling probabilities; typical outputs are statistical distributions of frequency, duration, coverage, and sampling probability; the strength is statistical representativeness; the limitation is that conclusions are conditional on the chosen environment pool and diagnostic definitions (Moore 2024). The eighth subcategory is near-surface diagnostics and post-processing studies, which focus on post-processing quantities such as peak factors, 50 m to 10 m diagnostic relationships, and swath construction to provide source-bounded examples for engineering use, but all such ratios are configuration-specific model-internal diagnostics rather than engineering conversion coefficients (Moore 2024; Orf et al. 2012).

These eight subcategories present a clear hierarchy in PETW relevance. Wet full-storm microphysics simulations and ensemble and synthetic-network studies are the closest to PETW needs for near-surface wind distributions under event conditions and sampling statistics, because they self-consistently represent the complete chain from convection to outflow and provide footprint

and network experiments that can serve as a sampling interface; high-resolution three-dimensional near-surface and circumferential studies provide source-bounded near-surface peak structure and peak factors and are key to understanding the relationship between model diagnostics and engineering gusts; dry or prescribed cooling-source and scaling-law studies provide mechanistic and controlling-parameter insight but, because they sacrifice real microphysics, should not be directly used for event-level near-surface wind; collision and downburst-line studies warn that isolated-profile assumptions fail in multi-downdraft interaction regions, which has cautionary significance for corridor-type structural exposure assessment; event-paired and WRF-linked studies connect real events with mechanism attribution but are limited by resolution and boundary settings. Quantitative results from different subcategories should not be migrated between one another; for example, a dry cooling-source peak factor cannot be directly applied to a wet microphysics event, and collision amplification cannot be applied to isolated outflow. This non-transferability is a principle that must be upheld in the subsequent synthesis of research progress by scientific purpose. This classification provides the framework for synthesizing research progress by scientific purpose in the next section.

6 Progress in CM1 Downburst Simulation Research

Research progress in CM1 downburst studies should be synthesized by scientific purpose rather than listed paper by paper. Centering on the core purpose of near-surface wind spatiotemporal distribution, Moore (2024) established the most systematic statistical characterization to date with a 97-member idealized ensemble; its purpose, methods, results, and limitations constitute the dedicated case study in Section 8 of this study, and here its contribution is first positioned within the overall research-progress context before the remaining nine direct CM1 records are treated individually. Regarding statistical representativeness, Moore's ensemble shows that under weakly forced wet thunderstorm wind-damage-report environments, 35% of simulations produce severe wind, the mean ratio of 50 m to 10 m diagnostics is 1.97, severe-wind coverage is on the order of O(10) km^2, the mean duration of exceeding 26 m/s at any point is approximately 5.2 min, and the probability of a typical mesoscale network sampling severe wind is approximately 0.7% (Moore 2024). These values are internal statistics of the ensemble under the selected environment and fixed configuration and provide a statistical background for the remaining single-simulation studies.

Centering on the purpose of near-surface fine structure and peak factors, two high-resolution three-dimensional studies provide key source-bounded results. The study simulating a microphysics-driven downburst-producing thunderstorm at extremely high resolution (approximately 20 m horizontal, approximately 5 m near-surface vertical) finds that the strongest near-surface wind occurs at near-surface heights (such as approximately 19 m and 38 m AGL) and can exceed 35 m/s, and circumferential analysis yields a peak factor of approximately 1.91 (to 100 m AGL) relating circumferentially averaged radial velocity to peak instantaneous radial velocity (Orf et al. 2012). The significance of this peak factor is that it indicates circumferential or temporal averaging within this configuration smooths out instantaneous local enhancements; 1.91 can only serve as a

source-bounded example from this single simulation and cannot serve as a load adjustment coefficient or a 3 s gust factor; the result is from a single simulation, the warm-bubble initial parameters were determined by undisclosed trial and error, it cannot be claimed that all downbursts share this spatiotemporal pattern, and the lack of a published namelist makes precise reproduction difficult. The study performing circumferential analysis on a simulated three-dimensional downburst to assess the effect of the axisymmetric approximation further shows that circumferential averaging preserves the maximum-wind region but smooths out horizontal ring-vortex and local turbulent structure, and in the strongest-wind region (radius less than or equal to 2000 m) the ratio of peak radial wind to circumferentially averaged radial wind is approximately 1.936 (Orf et al. 2014). This result has direct implications for the practice of using axisymmetric density-perturbation models or impinging jets as engineering outflow representations: the axisymmetric approximation, while preserving the location and magnitude of the maximum-wind region, systematically erases local turbulent enhancement; this ratio of approximately 1.936 indicates that circumferential averaging smooths instantaneous local enhancement, and it can only serve as a source-bounded example under a single modified dry downburst sounding in a controlled state; it cannot serve as a load coefficient or engineering gust factor, and it is not applicable to wet downbursts or strong-shear conditions. These two peak factors (approximately 1.91 and approximately 1.936) are numerically close and provide a consistent source-bounded example for the relationship between near-surface peaks and means, but both are configuration-specific, cannot be universalized, and cannot be conflated with an engineering 3 s gust factor.

Centering on the purpose of outflow evolution and scaling laws, several cooling-source and full-cloud studies provide complementary evidence. The study simulating the evolution and scaling of downburst-producing thunderstorms and comparing with an axisymmetric density-perturbation model finds that circumferentially averaged outflow velocity agrees with the axisymmetric density-perturbation model and that the sustained downdraft lasts at least 17 min rather than being a simple isolated pulse, indicating that a real downburst-producing thunderstorm can maintain strong outflow for several times the duration of a single pulse; this corrects the engineering habit of assuming short-pulse outflow duration; however, the URANS comparison model fails to reproduce the peak radial-wind height of CM1, the simulation itself contains complex interaction with a neighboring downdraft and is not fully axisymmetric, and the specific details of subgrid turbulence and microphysics scheme are NR (Oreskovic et al. 2018a). The parametric study using CM1 r18 LES with an idealized volumetric cooling source employs a constant 10 m horizontal grid with a vertical grid stretched from 1 m near the surface, sixth-order explicit horizontal advection, and a dry idealized cosine-squared space-time cooling function, and finds that the peak cooling rate significantly changes peak outflow speed within the tested matrix, but there is no universal power law outside the tested matrix; the Lundgren et al. (1992) constant-density droplet-release scaling successfully collapses the time- and space-varying cooling-source dynamics, indicating that cooling-source outflow possesses a scalable dynamical structure in non-dimensional variables (Oreskovic et al. 2018b). The engineering value of this scaling collapse is that it allows a family of cooling-source outflows to be summarized with a small number of non-dimensional parameters, but its domain of validity is strictly limited to the tested matrix and is premised on dry cooling

sources, and it cannot be extrapolated to wet events with real precipitation drag. The study testing Lundgren scaling, lapse-rate sensitivity, and vortex evolution with a CM1 cm1r19.10 cooling source and comparing with a full-cloud CM1 (time step 0.05 s, 10 m grid, 0.622 factor, 10% RMS band) finds that multiple normalized velocity and vortex diagnostics collapse within the reported error band, but the lowest-cooling case and part of the full-cloud comparison do not collapse, and reducing the lapse rate weakens and delays the dry forced-cooling outflow (Kondo 2021). These exceptions indicate that the cooling-source surrogate does not reproduce full-cloud dynamics at all intensities and environments, and the reported specific parameters (0.622 factor, 10 m grid, 0.05 s, 10% RMS band) are implementation details of that study, not universal PETW acceptance thresholds. The full-scale downburst parametric study employs a 1 to 50 m stretched vertical grid and Deardorff TKE closure, diagnoses that highly symmetric radial outflow under odd-order advection triggers a non-physical clefting artifact on the Cartesian grid, and eliminates it with a sixth-order explicit horizontal scheme combined with tuned diffusion, while also turning off the air-ground heat flux to isolate the dynamical process (Oreskovic 2016). This numerical-scheme recommendation is configuration-specific, indicating that even within the same model, the choice of grid geometry and advection scheme can introduce artifacts that require dedicated diagnosis, and default settings cannot be assumed safe. These studies collectively demonstrate that cooling-source surrogates within their tested configurations can reveal scaling relationships and numerical effects but do not establish reproduction of full-microphysics wind fields, and their reported implementation details are all configuration-specific rather than universal PETW acceptance thresholds.

Centering on the purpose of method comparison and event mechanisms, two studies provide boundary-type conclusions. The study improving CM1 cooling-source modeling for wind engineering and comparing with an impinging jet, employing quarter-domain mirror-symmetry boundaries and a thermodynamically formulated cooling function, finds that the CM1 cooling source in the tested configuration produces a near-surface flow topology significantly different from that of the impinging jet, with a non-dimensional vorticity-mapping deviation of at least 55%, and that the ring-vortex aspect ratio and vortex height are strongly correlated with surface roughness length whereas the impinging jet is essentially roughness-insensitive (Vermeire et al. 2011a); the study deliberately omitted the thermal inversion and has not published the exact namelist, time step, or custom cooling-function source modification, so it supports a topology difference in the tested configuration, not a verdict on impinging-jet similarity validity. The study analyzing the 9 June 2020 Akron macroburst event, combining an event-type WRF simulation with an independent horizontally homogeneous CM1 experiment (event sounding, warm-bubble trigger, Thompson microphysics, free-slip boundary, approximately 2.25 h integration, parcel analysis), finds that in a strongly sheared, stable post-frontal environment gravity-wave amplification and breaking contribute to intense surface winds, and parcel analysis traces the source and acceleration path of the strong-wind air parcels, indicating that in the specific environment of this event, besides the classic evaporative cooling and precipitation drag, gravity-wave breaking constitutes an additional mechanism for intense near-surface wind (Schumacher et al. 2023). The boundary of this finding is that the free-slip setting significantly overestimates extreme winds, the

WRF companion 4 km domain fails to resolve the correct wind magnitude and mechanism, and therefore the study supports a configuration-specific mechanism claim; it cannot claim that the observed event and CM1 peak winds match, nor can gravity-wave breaking be treated as a universal primary driver for all macrobursts. High-resolution observations of a destructive macroburst provide an observational counterpart for such mechanism diagnosis, but the comparison between observations and simulations remains at the mechanism level rather than pointwise quantitative validation (Childs et al. 2021). Centering on the purpose of outflow interaction, the study resolving downburst-line outflow and its interactions with CM1 LES finds that complex downdraft interaction can raise the peak maximum wind by up to 1.55 times relative to an isolated downburst and enlarges the damaging-wind footprint, the immediate collision region severely disrupts the primary ring vortex, and outflow profiles depart from standard isolated profiles (Vermeire et al. 2011b). The engineering implication of this collision amplification is that, within this idealized cooling-source collision configuration, the near-surface wind assumption based on isolated downbursts will systematically underestimate peak wind and damaging-wind footprint; however, the study uses an idealized thermal cooling source to stand in for microphysics, contains no ambient boundary-layer wind or strong tropospheric shear, and a real event with microphysics and shear may take this result only as a warning, not as a directly applicable 1.55 times amplification; the conclusion that classic isolated empirical profiles are no longer valid in collision regions is a robust cautionary finding. Synthesizing these ten direct CM1 records, it can be seen that CM1 has made specific contributions in the directions of near-surface structure, peak factors, scaling laws, numerical schemes, event mechanisms, and outflow interaction, but every quantitative result is strictly delimited to its single simulation or tested matrix and fixed configuration, and cross-environment or cross-subcategory migration is explicitly prohibited in each source. The remaining cited sources in the frozen corpus (predecessor models, observations, WRF events, physical experiments, and mechanism background) serve explicit functions in the method-history, validation, and mechanism-background sections and have not been forced into the research-progress narrative.

7 Initial Conditions, Physical and Numerical Settings, and Reproducibility

The credibility of CM1 downburst studies depends to a large extent on the transparency of initial conditions and numerical settings, and reported configuration, runnable reproduction package, and validated benchmark must be strictly distinguished as three non-equivalent tiers. At the initial-condition level, Moore's environment-screening workflow provides the most complete disclosure to date: starting from 49181 wind-damage local storm reports during 2018-01-01 to 2021-08-31 from NOAA Storm Data, for each report the nearest grid point and time of the RUC/RAP-based, hourly, 40 km SPC sfcOA is assigned and MUCAPE and EBWD are recorded; filtering by MUCAPE greater than 100 J/kg and EBWD less than 10.2 m/s (20 kt) retains 13069 reports, which based on prior work constitutes approximately 0.6% of all weakly forced thunderstorms; temperature, humidity, and wind profiles are collected at the corresponding grid points

with 25 mb vertical resolution, a mean profile is compiled yielding a surface-to-3 km lapse rate of approximately 5.0°C/km and a wet-downburst-type thermodynamic structure, and then reports are culled using a 0 to 3 km relative humidity standardized deviation (Z-score) cutoff of ±2 to remove 530 excessively moist or dry soundings, yielding 12539 candidates; random sampling without replacement and success criteria (convection generated, initial downburst inside domain, numerically stable full 2 h integration) with spatiotemporal deduplication finally yield 97 valid simulations (Moore 2024). This workflow demonstrates that the 97 members are a sample screened from a broader wind-damage-report environment pool and do not define a universal wet-downburst threshold, and the composite sounding (mixed-layer LCL below 800 mb, 0 to 3 km lapse rate approximately 7.4°C/km, layer-mean relative humidity approximately 68%, mean EBWD approximately 6.2 m/s) represents a subset of weakly forced wet-downburst environments.

Assessment of environmental representativeness further depends on scrutiny of the analysis-field sounding source. SPC sfcOA is RUC/RAP-based, with 40 km horizontal resolution and hourly output; its advantage is broad coverage and spatiotemporal continuity, making it convenient for assigning environmental conditions to each of the many wind-damage reports, but as an analysis field rather than an observed sounding it contains systematic biases; the author notes that the environments in this study generally have high MUCAPE (exceeding 1000 J/kg), so the relative impact of analysis-field bias is small, and the bias between the custom EBWD and sfcOA EBWD is approximately 1.2 m/s, on the basis of which the author argues that this bias does not dominate the environmental classification (Moore 2024). The mean profile of the 13069 candidates used for dry-wet discrimination yields a surface-to-3 km lapse rate of approximately 5.0°C/km (far smaller than the 9.0 to 10.0°C/km of dry downbursts), surface-to-850 mb relative humidity of approximately 66% and 850-to-500 mb relative humidity of approximately 62%, a theta-e deficit of approximately 21.0 K, and MLCAPE of approximately 1214 J/kg; these quantities collectively position this environment pool as wet-downburst type rather than dry type (Moore 2024). Comparing the environmental distributions of the 12539 candidates and the 97 input soundings with existing wet-downburst environmental documentation, the interquartile ranges of MLCAPE, vertical totals, total totals, temperature-dewpoint deficit, and DCAPE fall within similar ranges supporting severe wind from weakly forced thunderstorm events, indicating that the environmental distribution of this ensemble has defensible representativeness for wet downbursts, but this representativeness is for a particular environment family, not for all wet-downburst events (Moore 2024). The 97 input soundings span May to October in time, 90% are concentrated in June to August, and 74% fall within 1800 to 2200 UTC, further indicating that they represent warm-season afternoon-to-evening weakly forced moist convection.

At the level of physics and numerical settings, each setting choice has physical and numerical significance for the results. The thermal perturbation amplitude and geometry of the warm bubble determine the initial updraft intensity and convective triggering timing, and Moore's sensitivity tests with 2 to 6 K thermal perturbation across five sets show variability small enough that the default 2 K is retained (Moore 2024); in cooling-source studies, the cooling-function shape, peak cooling rate, and spatiotemporal scale directly determine outflow intensity and scal-

ing relationships (Oreskovic et al. 2018b; Kondo 2021). The microphysics scheme determines the representation of hydrometeor phase change and precipitation drag, lapse rate and low-level humidity determine the efficiency of negative buoyancy generation, and deep-layer and low-level shear determine convective organization and outflow symmetry. Surface friction, roughness, land use, and lower-boundary type jointly determine near-surface wind attenuation, and the semi-slip lower boundary employed by Moore may produce overly high surface friction and bias 10 m wind low when turbulent eddies are not introduced at initialization (Moore 2024). Grid spacing, number of vertical levels, time step, output frequency, domain size, lateral boundaries, subgrid closure, advection order, and diffusion jointly determine the resolvability of peak wind and numerical artifacts, and Moore's grid sensitivity shows that increasing horizontal spacing, while only slightly lowering peak wind (2.4% to 12.8% between 0.1 and 0.5 km), produces stronger downbursts and larger outflow radii, suggesting that peak wind may be under-resolved (Moore 2024).

At the reproducibility level, a clear gap exists between reported configuration and a runnable reproduction package. Moore's data-availability statement indicates that the CM1 namelist and candidate sounding dataset are available upon request and that the CM1 code is publicly downloadable (Moore 2024), meaning that the configuration is sufficiently described but a runnable package is not directly provided with the paper; most other studies do not attach namelists, source modifications, or exact time steps with their papers, so their reproducibility fields are mostly NR. Cooling-source studies are particularly affected: their externally imposed cooling functions are often custom modifications to CM1 (such as specific cosine-squared ellipsoidal cooling sources), and exact reproduction requires implementing the source-code modification for that cooling function rather than using default settings alone (Oreskovic et al. 2018b; Kondo 2021; Vermeire et al. 2011a). Therefore, even if a study reports its grid, advection order, and time step, its runnable reproduction may still be blocked by unpublished source-code modifications and initial-perturbation details. A validated benchmark is an even higher threshold: the two-dimensional rising thermal bubble conservation benchmark and the deep moist convection resolution-requirement study provide configuration-specific conservation and resolution sensitivity (Bryan and Fritsch 2002; Bryan et al. 2003), not event-level peak-wind validation. Thus, in the PETW context, the reported configuration of any study cannot be automatically taken as a runnable package, a runnable package cannot be automatically taken as a validated benchmark, and this three-tier distinction (reported configuration, runnable reproduction package, validated benchmark) is the prerequisite for the subsequent PETW interface discussion and the baseline framework for evaluating the transferability of any quantitative result.

8 The Moore (2024) Core Case

Moore (2024) (Zotero key JFT4WRDC, DOI 10.1175/WAF-D-23-0164.1) uses a 97-member high-resolution idealized single-cell downburst simulation to make statistical inferences about the near-surface wind spatiotemporal distribution of weakly forced thunderstorm wet downbursts; it is the central methods-results-limitations spine and writing-tone reference of this study. Its research

question is that weakly forced thunderstorms are common during the warm season in the eastern and southeastern United States, their wet-downburst outflow can damage trees, power lines, and property, yet the frequency of severe downbursts, the spatial extent, and the temporal duration remain unclear, and whether modern observation networks can reliably observe them is unknown; to answer these questions in a statistical sense requires an ensemble covering diverse damaging environments rather than a single case (Moore 2024). The data flow has been described in Section 7: starting from 49181 wind-damage reports, environmental filtering yields 13069, dry-wet filtering yields 12539 candidates, and random sampling with success criteria and deduplication ultimately yields 97 valid simulations.

In model configuration, Moore employs CM1 Release 20 version 3, an 80 km $\times$ 80 km horizontal domain, 0.1 km horizontal spacing, 15 km model depth, 84 vertical levels, 0.1 km vertical spacing below 2 km, with the lowest model level at approximately 50 m AGL, a time step of 1 s, data output every 60 s, and full 2 h integration (Moore 2024). Each storm is triggered by a single positive 2 K warm bubble in a horizontally homogeneous environment, with horizontal width 10 km, vertical depth 1.5 km, starting height 0.5 km, located at the domain center and held fixed across all simulations. Additional settings are sixth-order horizontal advection, fifth-order vertical advection, sixth-order diffusion, Deardorff TKE subgrid turbulence, semislip lower boundary, open radiative lateral boundaries, free slip upper boundary, radiation scheme None, MM5 similarity-theory surface layer, and low-intensity residential land use; microphysics is Morrison double-moment predicting mixing ratio and number concentration of rain, snow, graupel, and hail. Figure 2 shows the simulated air-mass thunderstorm life cycle and its morphological comparison with the 12 July 2018 observed downburst, and Figure 3 shows the instantaneous 50 m wind, potential-temperature perturbation and reflectivity cross section, vertical wind profile, and 10/50 m wind-speed time series at model time 2220 s; together they present the physical chain from convective triggering to cold-pool expansion, outflow asymmetry, and the low-level wind maximum. As an internal plausibility check, all simulations produce surface precipitation, with mean maximum precipitation of 2.56 cm, nearly all simulations peak in intensity within the first 3600 s, the strongest downburst outflow is asymmetric with the strongest wind on the downwind side and a classic rotor on the upwind edge of the cold pool, the strongest wind within the gust front is at approximately 50 m AGL while above 250 m it drops rapidly below 10 m/s, and the remaining 96 simulations also have peak radial velocities at 50 or 100 m AGL (Moore 2024).

The definitional discipline of near-surface diagnostics is key to Moore's methodology. CM1 records the maximum horizontal wind at the lowest horizontal velocity level at every time step (1 s) and writes it as a maximum-wind swath grid every output step (60 s); because of the 100 m vertical spacing, this level is at approximately 50 m AGL; instantaneous surface diagnostic variables are also output every 60 s and include 10 m U/V wind, with the 10 m maximum-wind grid constructed from the maximum 10 m wind at each output step (Moore 2024). Across all simulations, 10 m wind is weaker than 50 m wind, with the mean 50-to-10 m wind-speed ratio of 1.97, consistent with existing high-resolution numerical studies and observations. Using the 10 m maximum wind has a limitation: the diagnostic surface variables are instantaneous values

at each output step rather than maxima or averages across all model steps, so the 10 m peak may fall between output steps and be missed, manifesting as a ripple effect in the 10 m wind field, and Figure 5 compares the differences between the 50 m and 10 m maximum-wind swaths. Because the semislip lower boundary may produce overly high surface friction and bias 10 m wind low when turbulent eddies are not introduced at initialization (citing Markowski and Bryan 2016), Moore uses 50 m wind for analysis and interprets it as a 10 m gust proxy rather than a sustained wind, citing precedents of using 80 m wind as a 10 m gust proxy and using 50 m gust for structural and transmission-tower peak-wind assessment (Moore 2024). The update and sampling definitions of this 50 m proxy are both different from those of an engineering 3 s gust, and the 1.97 ratio is a model-internal diagnostic ratio, not an engineering conversion coefficient.

In ensemble statistics, 34 simulations reach or exceed the NWS severe threshold, nearly all simulations experience their peak within approximately 1000 s after initial downburst touchdown, the mean (not necessarily continuous) duration of wind exceeding 26 m/s at any point in the domain is 5.2 min (minimum 1 min, maximum 17 min), and the mean total duration exceeding 20 m/s is approximately 25 min (Moore 2024). Using the end-of-simulation 50 m maximum-wind swath for each simulation as the spatial analysis and masking below the field median to isolate the downburst wind field, 80% of simulations contain wind of 20 m/s or above, 53% contain 24 m/s or above, 35% produce severe wind, peak wind of 37 m/s occurs in a single simulation only, and the 50 m wind PDF is gamma-distributed (15 m/s at the 90th percentile, 21 m/s at the 99th, 26 m/s at the 99.9th). Spatial coverage decreases with threshold, the median coverage at 26 m/s falls to approximately 100 km^2, and only 8 of 34 (23%) severe-wind swaths exceed 10^1 km^2; Figure 4 shows the coverage-area distribution by wind threshold. Based on the eight largest-severe-extent simulations suggesting that a maximum station spacing on the order of O(3) km is needed to sample wind exceeding 26 m/s, Moore constructs MESO, MICRO, and MIST synthetic observation networks (mean station spacings of 33.4 km, 5.2 km, and 2.3 km respectively), builds 100 networks per swath for each of the 34 severe-producing swaths, and obtains the mean sampling probability of severe wind across any scale: MIST 43%, MICRO 18%, MESO 0.7%; from a Nyquist-type analogy the minimum resolvable area is defined as

$$\mathrm{MRA} = (2S)^2, \tag{1}$$

where S is the minimum station spacing of the observation network, yielding MIST, MICRO, and MESO values of 21.2, 108.2, and 4462.2 km^2 respectively; the MESO value of 4462.2 km^2 is a sampling-scale reference within Moore's synthetic-network framework, not a guaranteed detection area of any real mesoscale network (Moore 2024). The construction of the synthetic-network experiment follows an explicit sampling geometry. Moore uses the formula for mean station spacing of a regular network given by Koch et al. (1983) as a basis, simplified to relate station count to domain area and mean station spacing (station count approximately equals domain area divided by station spacing squared), places stations randomly within the 80 km × 80 km domain with a minimum-spacing constraint, constructs 100 networks per swath for each of the 34 severe-producing swaths for MESO, MICRO, and MIST, yielding 3400 networks per class and 10200 total, with a hit defined as at least one station in the network reaching the specified threshold. The

sampling probabilities of the three network classes exhibit two robust relationships: for a given threshold coverage, sampling probability decreases with increasing station spacing (at 26 m/s with 10^1 km^2 coverage: MIST approximately 80%, MICRO approximately 20%, MESO less than 5%); for a given station spacing, sampling probability increases with increasing coverage area and follows a logistic shape. The spatial scale at which 50% sampling probability is reached increases with station spacing: MIST (2.3 km) corresponds to approximately 10^0 km^2, MICRO (5.2 km) to approximately 10^1 km^2, and MESO (33.4 km) to approximately 10^2 km^2. Substituting the minimum resolvable area into the fitted logistic relationship, the sampling probabilities of MIST and MICRO reach approximately 98% and approximately 100% respectively. Regular-grid station placement slightly outperforms random placement: regular MIST networks sample wind of at least 22 m/s in every downburst wind field and capture 50% of severe-wind swaths; regular MICRO captures 21%; and regular MESO captures 6%. These results collectively support the PETW sampling-interface discussion, but their conclusions are strictly conditional on idealized swaths, the 50 m diagnostic, specified station spacings, and the severe subset, and cannot be treated as a universal detection guarantee for real networks.

Moore derives three principal conclusions from the ensemble and network experiments. First, the CM1 simulations suggest that given a weakly forced thunderstorm wind-damage report, the frequency of damaging downbursts containing severe wind is approximately 35%; combined with the approximately 0.6% subset from prior work, severe weakly forced thunderstorms constitute only approximately 0.2% of all weakly forced thunderstorms, and the true value may be lower. Second, the simulated-network results suggest that a mean station spacing of approximately 1.58 km or smaller is needed to reliably sample the severe-wind swath of weakly forced thunderstorm downbursts. This 1.58 km value is only a simulated-network result and not an operational specification; few operational networks meet this density, and the probability of a typical mesoscale network detecting a gust exceeding 26 m/s is only approximately 0.7%. Third, sub-severe winds are more frequent and have larger footprints, so most tree damage caused by pulse convection may originate from sub-severe wind rather than severe wind; canopy-level winds of 10 to 20 m/s can already damage trees. These three conclusions raise a discussion of the operational practice of verifying severe warnings based on tree-damage reports and also provide a direct basis for the subsequent roadmap (Moore 2024).

Regarding validation and limitations, Moore uses the 12 July 2018 observed weakly forced thunderstorm at the Mississippi and Tennessee border (KNQA 0.5° reflectivity) as a morphological comparison; this is morphological plausibility, not event-matched quantitative validation (Moore 2024). Using spatial coverage as an example, Moore cites the national mean severe thunderstorm warning area of approximately 1802 km^2 to illustrate that a severe-wind swath on the order of 10^1 km^2 corresponds to only approximately 0.9% of the warning area, and the maximum severe coverage (on the order of 10^2 km^2) corresponds to only approximately 9.2%; this comparison illustrates the sparseness of the severe downburst wind footprint relative to the warning area but serves only as an illustration, not as an accuracy statement (Moore 2024). The author explicitly lists the following key limitations: increasing grid spacing, while affecting peak wind by only approximately 2.4%, produces stronger downbursts and larger outflow radii, suggesting that

peak wind may be under-resolved; terrain is not considered, and terrain can significantly affect near-surface wind; the sensitivity of the semislip lower boundary and friction parameterization is not explicitly tested, and this may affect low-level velocity. The author judges that no single factor is sufficient to significantly change the overall conclusions, but these three self-declared limitations should not be suppressed. In the PETW context, Moore provides a description of reproducible assets (namelist and soundings available upon request, code publicly downloadable) and a series of NR fields (time-integration scheme, acoustic treatment, Coriolis setting, event-level quantitative validation, 10 m 3 s gust and `Vhat_3s` conversion, terrain and friction sensitivity). The 35% should not be treated as a universal real-world severe-wind frequency, the 0.7% should not be treated as a universal mesoscale-network detection probability, and 50 m wind should not be treated as an engineering 10 m 3 s gust or `Vhat_3s`; these boundaries must be maintained before Moore's results enter PETW.

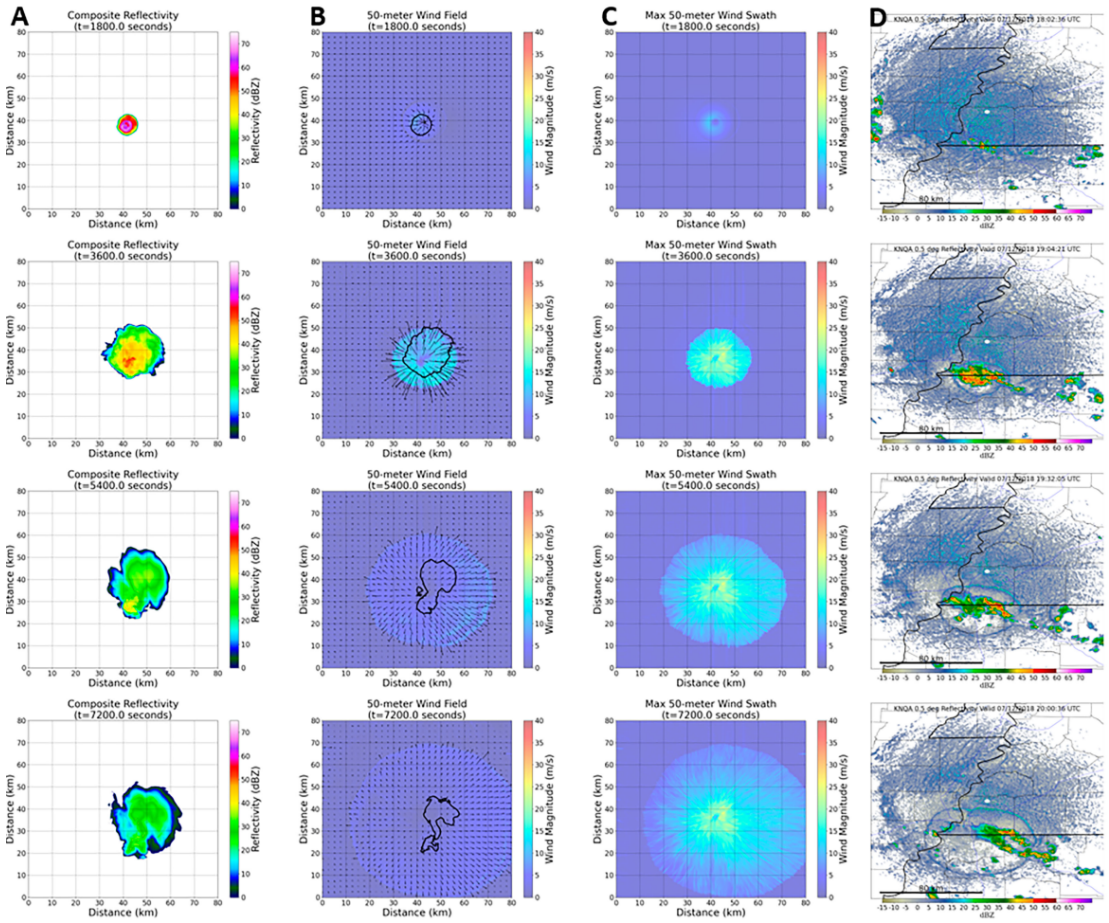


Figure 2: Life cycle of the CM1-simulated air-mass thunderstorm and its morphological comparison with the 12 July 2018 observed downburst, from Moore (2024) (Zotero key JFT4WRDC), original Figure 4, located on page 9 of that paper's PDF. Columns from left to right are composite reflectivity, instantaneous 50 m wind, 1800/3600/5400/7200 s accumulated 50 m wind, and KNQA 0.5° observed reflectivity. The scientific function is to show the morphological plausibility comparison between the simulated life cycle and the observation, demonstrating morphological plausibility rather than event-matched quantitative validation. This is an internal-review crop with status `internal_review_only_permission_or_redraw_required_before_publication`; copyright permission must be obtained or the figure redrawn before formal publication, and this study does not imply that permission has been obtained.

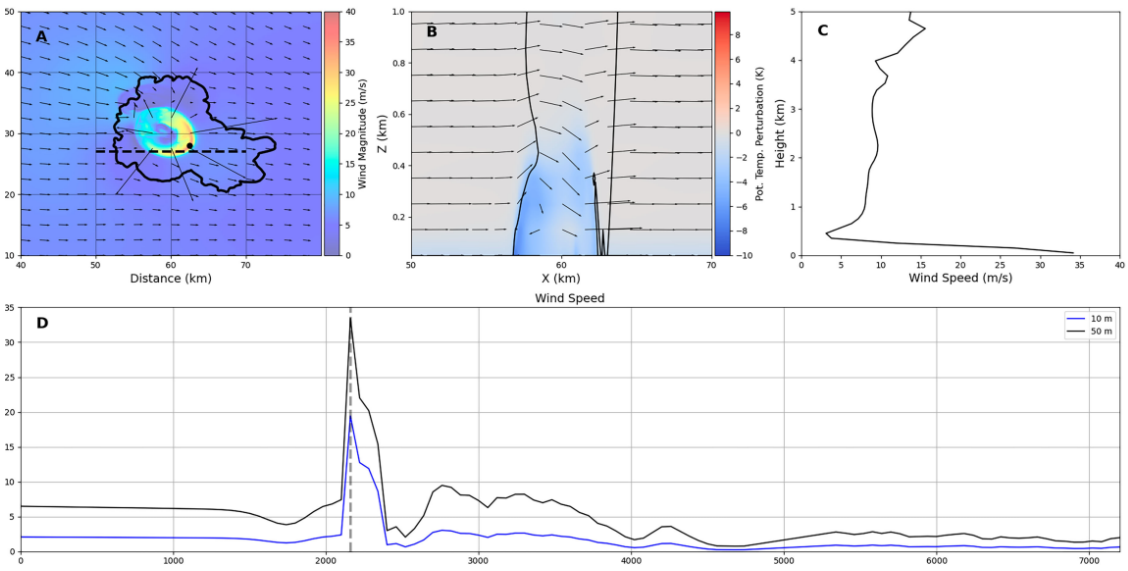


Figure 3: Instantaneous 50 m wind, potential-temperature perturbation and 25 dBZ reflectivity cross section, vertical wind profile, and 10/50 m wind-speed time series at model time 2220 s, from Moore (2024) (Zotero key JFT4WRDC), original Figure 5, located on page 10 of that paper's PDF. The scientific function is to connect the CM1 physical structure, low-level wind maximum, and 50/10 m diagnostic definitions. This is an internal-review crop with status `internal_review_only_permission_or_redraw_required_before_publication`; copyright permission must be obtained or the figure redrawn before formal publication, and this study does not imply that permission has been obtained.

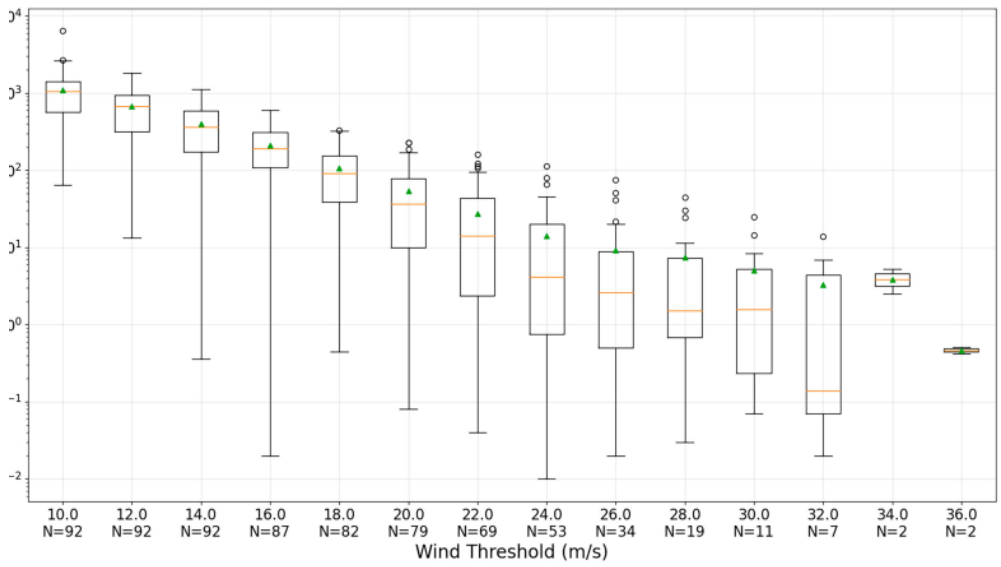


Figure 4: Distribution of 50 m wind coverage area (km^2) by wind threshold (m/s), with median shown as orange bars and mean as green triangles, from Moore (2024) (Zotero key JFT4WRDC), original Figure 10, located on page 13 of that paper's PDF. The scientific function is to connect threshold, footprint area, and point or network sampling probability. This is an internal-review crop with status `internal_review_only_permission_or_redraw_required_before_publication`; copyright permission must be obtained or the figure redrawn before formal publication, and this study does not imply that permission has been obtained.

9 Comparison of Simulations with Measurements, Physical Experiments, and Other Models

Assessing the credibility of CM1 downburst simulations requires distinguishing among five different validation tiers: qualitative morphological agreement, source-specific quantitative consistency, event reconstruction, laboratory similarity, and benchmark verification, avoiding the extrapolation of success at any one tier into universal accuracy. At the qualitative morphological tier, Moore's simulated life cycle morphologically agrees with the reflectivity evolution of the 12 July 2018 observed weakly forced thunderstorm, and the asymmetric strongest outflow, downwind-side enhancement, and rotor on the upwind edge of the cold pool are also consistent with existing experiments and observations (Moore 2024), but the author explicitly positions this as morphological plausibility rather than event-matched quantitative validation. At the source-specific quantitative tier, each individual study provides specific quantities within its configuration, such as peak wind, peak height, peak factor, and duration (for example, peak factors of approximately 1.91 and approximately 1.936, and a sustained downdraft of at least 17 min); these quantities are internally self-consistent within their respective simulations but, being limited to a single sounding and fixed configuration, do not constitute cross-event quantitative validation (Orf et al. 2012, 2014; Oreskovic et al. 2018a).

At the event-reconstruction tier, multiple studies along the WRF line perform regional simulations targeting real damaging events for mechanism diagnosis and near-surface wind reconstruction. Relevant work includes microburst detection and effective-resolution studies (Bolgiani et al. 2020b), high spatiotemporal-resolution mesoscale simulation (Bolgiani et al. 2020a), analyses of tower-collapse events and destructive-wind thunderstorm mechanisms (Li et al. 2023; Liu et al. 2023), testing of a gust-front head-collapse scheme (Lompar et al. 2018), and investigation of weather conditions during a bridge collapse (Burlando et al. 2020). These event simulations can be used to reconstruct or diagnose the spatiotemporal background and mechanisms of events, but their coarser grids typically cannot correctly resolve near-surface peak winds at the burst-swath scale, and the Akron event comparison study shows that the 4 km domain fails to resolve the correct wind magnitude and mechanism (Schumacher et al. 2023). The value of event reconstruction is in linking simulations to specific observed events, enabling qualitative to semi-quantitative comparison at the synoptic-scale environmental, triggering, and evolution-path levels; its limitation is that even when the synoptic-scale background is correctly reproduced, near-surface peak wind and fine outflow structure may still deviate from observations because of resolution, boundary, and friction settings, so event reconstruction can establish the plausibility of a mechanism but cannot by itself establish the quantitative accuracy of near-surface peak wind. This point is complementary with idealized ensembles: idealized ensembles trade event correspondence for statistical representativeness, event reconstruction provides synoptic-scale and mechanism correspondence but does not automatically obtain quantitative credibility for near-surface peaks, and neither can alone provide event-matched pointwise quantitative validation. At the laboratory-similarity tier, impinging-jet experiments (including high-resolution measurements at the WindEEE Dome) provide controlled data on outflow kinematics and roughness effects (Sengupta and Sarkar 2008;

Canepa et al. 2022; Zhang et al. 2013; Lin et al. 2007), but the non-dimensional vorticity deviation between impinging jets and cooling sources in near-surface flow topology is at least 55%, indicating that laboratory similarity cannot automatically equate to CM1 outflow (Vermeire et al. 2011a).

At the observational-constraint tier, multiple observational methods provide target quantities and comparison references for validation. The observations of Colorado microburst outflow structure and life cycle systematically characterize the evolution of a microburst from touchdown to dissipation and the shallowness of outflow (Hjelmfelt 1988); Doppler and rawinsonde observations of the gust-front life cycle reveal head structure and propagation (Wakimoto 1982); joint measurements from the TTUKA mobile Doppler radar and StickNet provide high spatiotemporal-resolution sampling of outflow wind in the low levels and are explicitly used for wind-engineering applications (Gunter 2015); high-resolution full-scale thunderstorm outflow measurements provide the turbulence and gust characteristics of near-surface wind (Gunter and Schroeder 2015); combined Doppler profiler and near-ground measurements characterize the vertical structure of downburst wind (He et al. 2024); high-resolution observations of a destructive macroburst provide event-level constraints for macroburst near-surface wind (Childs et al. 2021); and Livorno event field data and weather scenario analysis, an all-digital phased-array radar downburst-precursor comparison framework, and video-based kinematics assessment expand the observational methods and precursor identification from different angles (Burlando et al. 2017; Pearson et al. 2023; Romanic and Vavatsikos 2024). These observations collectively characterize the peak magnitude, peak height, radial and circumferential structure, outflow radius, duration, pressure and temperature signals, and near-surface wind profiles of outflow. These observations establish the target quantities for validation (peak magnitude, maximum-wind height, radial or circumferential structure, outflow radius, duration, pressure and temperature signals, reflectivity or hydrometeor structure, footprint area, time series, and spatial sampling), but most observations, limited by lowest elevation angle, station spacing, and near-surface sampling capability, are unable to provide event-matched pointwise quantitative tolerances for simulations. At the benchmark-verification tier, the two-dimensional rising thermal bubble conservation benchmark and the deep moist convection resolution-requirement study provide configuration-specific numerical conservation and resolution sensitivity rather than validation of downburst near-surface peak wind (Bryan and Fritsch 2002; Bryan et al. 2003). Synthesizing across five tiers, current CM1 downburst research reaches the level of morphological plausibility plus source-specific quantitative consistency; event-matched pointwise quantitative validation and a unified tolerance do not yet exist; therefore this study does not state any single universal accuracy percentage, and where matched validation or tolerance is absent it states so directly.

10 Near-Surface Wind-Field Issues

The near-surface wind field is the most PETW-relevant scientific element, and its core issue is that multiple systematic mismatches exist between CM1 near-surface diagnostics and an engineering 10 m 3 s gust, each of which must be explicitly identified. The first mismatch is height: Moore's

lowest model level is at approximately 50 m AGL, and its 50 m maximum wind differs from the engineering reference height of 10 m; although a separate 10 m diagnostic wind exists, the physical height difference between the two produces a mean 50-to-10 m ratio of 1.97 (Moore 2024). The second mismatch is temporal sampling: the 50 m swath is updated every 1 s and written every 60 s, while the 10 m diagnostic is a 60 s instantaneous value; the 10 m peak may fall between output steps and be missed, producing the ripple effect, so the 10 m maximum wind is not the true 10 m peak, and the engineering 3 s gust requires a sliding extreme over a specific averaging period that corresponds to neither (Moore 2024). The third mismatch is spatial sampling: model output consists of gridpoint quantities, whereas engineering use often requires station-point or specific spatial-average values, and the gridpoint-versus-station sampling and vertical-interpolation methods alter the near-surface wind statistics. The gridpoint maximum-wind swath records the maximum wind experienced at every grid point over the entire simulation, an Eulerian spatial envelope, while a single station records the time-series extreme at that fixed point, and the statistical meanings differ; directly interpreting the gridpoint swath as the wind a station would observe systematically overestimates the sampling result of a fixed observation point, and Moore's synthetic-network experiment is designed precisely to quantify this difference. Regarding vertical interpolation, the engineering reference height (such as 10 m) is often below the lowest model level (such as 50 m), and interpolation from the model level to the engineering height must rely on a near-surface wind-profile assumption, but the near-surface profile of downburst outflow is itself highly non-logarithmic and changes rapidly with gust-front position, so any fixed profile assumption introduces error.

The fourth mismatch is the vertical structure of the low-level wind maximum: the strongest wind within the gust front is at approximately 50 or 100 m AGL, above 250 m it drops rapidly below 10 m/s, indicating that the outflow wind is shallow, and any single fixed diagnostic height captures only one cross section of the vertical profile and cannot represent the near-surface gradient required by engineering (Moore 2024). The fifth mismatch is horizontal asymmetry and transience: the gust-front head, ring vortex, and rotor cause the strongest wind to be displaced to the downwind side and to have directional structure, and the outflow direction changes rapidly as the gust front passes, which contradicts the directional stability assumption of stationary boundary-layer wind (Moore 2024). The sixth mismatch is outflow interaction: when multiple downdrafts collide, the peak maximum wind can be raised by up to 1.55 times and the damaging-wind footprint enlarged, and collision-region outflow profiles depart strongly from isolated empirical shapes, causing the near-surface wind assumption based on isolated profiles to fail in these regions (Vermeire et al. 2011b). Figure 5 visually presents the differences in spatial morphology and amplitude between the 50 m and 10 m maximum-wind swaths and is key to understanding the height and temporal mismatches described above.

The seventh mismatch is surface roughness and friction: the semislip lower boundary employed by Moore may produce overly high surface friction and bias 10 m wind low when turbulent eddies are not introduced at initialization (citing Markowski and Bryan 2016), and the roughness effect of a real engineering site needs to be explicitly modeled; at present this friction bias is not quantitatively corrected within the corpus (Moore 2024). The direction of this bias is important:

if the semislip friction biases near-surface wind low, then using the model 10 m wind directly as engineering input may underestimate the near-surface wind input, whereas using the 50 m proxy, being at a greater height, may not correspond to the 10 m engineering gust; the effect of this error on loading is not quantified in the current corpus and cannot be directly translated into structural loading conclusions; both directions of error must be explicitly modeled rather than offset with a single coefficient. The cooling-source and impinging-jet comparison study further shows that the ring-vortex aspect ratio and vortex height are strongly correlated with surface roughness length, indicating that near-surface flow topology is sensitive to roughness and that roughness is not a negligible second-order factor (Vermeire et al. 2011a). The eighth mismatch is terrain neglect: Moore's idealized simulation does not consider terrain, and terrain can significantly alter near-surface wind acceleration, separation, and local enhancement; this is the author's self-declared limitation (Moore 2024). For transmission corridors over complex terrain, terrain-induced local acceleration may superpose with downburst outflow, causing near-surface wind at certain towers to be far higher than the flat-terrain assumption; this effect is entirely omitted in idealized simulations and must be supplemented with dedicated terrain-sensitivity experiments. The ninth mismatch is turbulence and nonstationarity: downburst outflow is inherently nonstationary, its wind-speed time series exhibits rapid onset and subsequent decay with oscillatory surges as the gust front passes, the mean itself varies with time, and therefore gust-factor, peak-factor, and turbulence-intensity definitions premised on stationarity cannot be directly applied (Moore 2024). In engineering practice, short-window sliding averages are commonly used to define stationary segments, but the characteristic time scale of downburst outflow is on the order of minutes, so short as to make defining a stable stationary segment difficult, and extracting engineering-meaningful gust statistics from model time series itself becomes a methodological problem requiring dedicated treatment. The performance of subgrid turbulence closure (such as Deardorff TKE) near the surface further affects simulated turbulence intensity, and whether this closure is sufficient for downbursts has not been established, so the model-produced turbulence quantities carry closure-dependent uncertainty. The tenth mismatch is the relationship between peaks and means: high-resolution studies provide peak factors of approximately 1.91 and approximately 1.936 as source-bounded examples for the near-surface peak-to-mean relationship, but these factors are configuration-specific model-internal diagnostics and cannot serve as universal engineering gust factors (Orf et al. 2012, 2014). The eleventh mismatch is duration, footprint, and sampling detection: the mean duration of wind exceeding 26 m/s at any point is approximately 5.2 min (minimum 1 min, maximum 17 min), the mean total duration exceeding 20 m/s is approximately 25 min, severe-wind coverage is on the order of approximately 100 km^2, and only 8 of 34 severe-producing simulations have swaths exceeding 10^1 km^2; the typical mesoscale-network sampling probability is approximately 0.7% (Moore 2024). These statistics both characterize the spatiotemporal limitations of near-surface wind and demonstrate the systematic undersampling of near-surface peaks by observations: severe wind is both brief and localized, precisely the signal most easily missed by sparse observation networks, and this is the core motivation for Moore's use of simulation to supplement observations. For PETW, this means that any severe-wind probability estimated from historical observational frequency may be biased low due to undersampling, and the distributions given by simulation must be interpreted

conditional on their configuration boundaries. The twelfth mismatch is uncertainty propagation: each of the above mismatches introduces an independent uncertainty, and the conversion from model diagnostics to engineering gusts must propagate these uncertainties link by link rather than substituting a single coefficient. Comparing model diagnostics with observation heights and instrument limitations, Doppler radar, profilers, and StickNet each have their own height and sampling limitations (Gunter 2015; Gunter and Schroeder 2015; He et al. 2024), and therefore the near-surface wind comparison between model and observations must itself be conducted at matched height, averaging period, and spatial scale. In summary, no CM1 near-surface diagnostic is equivalent to `Vhat_3s`, the 1.97 ratio is a model-internal diagnostic ratio and not an engineering conversion coefficient, and every mismatch in the near-surface wind chain is an object that the PETW interface must explicitly model and constrain with uncertainty.

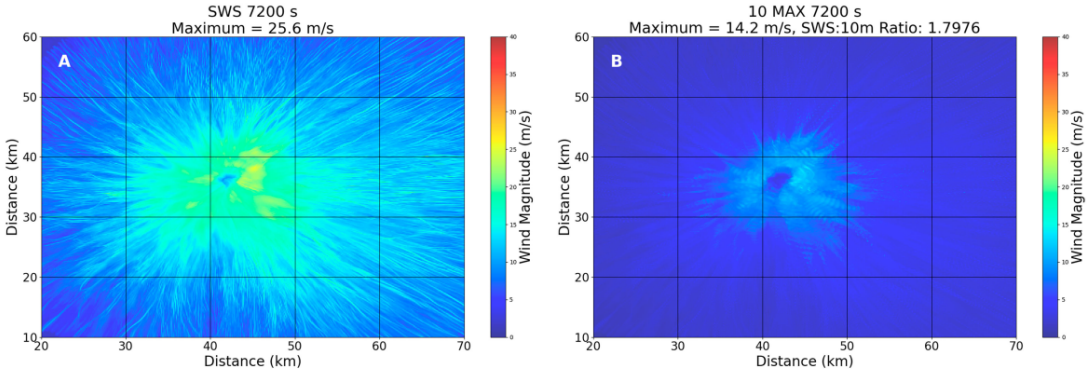


Figure 5: Comparison of the 50 m maximum-wind swath (a) and the 10 m maximum-wind swath (b) at the simulation end time of 7200 s, from Moore (2024) (Zotero key JFT4WRDC), original Figure 6, located on page 10 of that paper's PDF. The scientific function is to illustrate the difference in update and sampling definitions between the 50 m and 10 m diagnostics; neither is an engineering 3 s gust. This is an internal-review crop with status `internal_review_only_permission_or_redraw_required_before_publication`; copyright permission must be obtained or the figure redrawn before formal publication, and this study does not imply that permission has been obtained.

11 How CM1 Results Can Be Used by PETW

Using CM1 results for PETW requires an operable workflow that maintains unapproved status, with each interface stage explicitly identifying what can be provided, what transform is missing, what must be calibrated against, and what uncertainty must be propagated. Figure 6 presents this source-bounded path from event and mechanism scenario to engineering gusts as an original synthesized schematic. Interface stage A is event and mechanism scenario selection: a real event and mechanism background (such as macroburst observations and event-paired studies) define an environment family, but this step only bounds the environment and does not establish a simulation ground truth (Schumacher et al. 2023; Childs et al. 2021). Interface stage B is sounding source and conversion: the CM1 input sounding is derived from the event-conditional environment in an auditable manner; Moore's sfcOA derivation, 25 mb resolution, and dry-wet filtering workflow

provide a method reference for this step, but the sfcOA sounding is an analysis field, not an observed sounding, and its bias must be recorded (Moore 2024). Interface stage C is CM1 subcategory choice: the appropriate CM1 subcategory is selected according to mechanism (dry cooling source, wet full storm, collision or line, or event-paired) with a frozen namelist and run manifest attached; most studies currently have namelists that are available upon request or NR, so this step requires supplementing a runnable package before PETW reproduction can proceed (Moore 2024).

Interface stage D is raw output requirements: the required raw outputs include the three velocity components u, v, w, thermodynamic and pressure perturbation, hydrometeors or reflectivity proxy, coordinates, time series, surface variables, and the maximum-wind swath and selected three-dimensional snapshots, and the output frequency must be high enough to avoid missing peaks. Moore's experience shows that outputting instantaneous surface diagnostics at 60 s intervals misses 10 m peaks between output steps and produces the ripple effect, so if PETW requires reliable near-surface peaks, the output frequency must approach the model time step or employ a runtime sliding-extremum accumulation rather than taking extrema from sparse snapshots after the fact (Moore 2024). This output-strategy choice directly determines the credibility of all subsequent near-surface wind statistics and is a configuration decision that must be frozen before the run. Interface stage E is near-surface processing: from surface diagnostics, 10 m u and v wind components and 10 m maximum wind are constructed, and the model-level (such as 50 m) swath is separately preserved; the three quantities must not be equated, because their update and sampling definitions differ (Moore 2024). The near-surface processing stage should preserve all intermediate quantities (model-level wind, 10 m diagnostic wind, and the update frequency and sampling definition of each) and their metadata, so that downstream transforms can auditably trace the source and definition of every quantity rather than collapsing to a single near-surface wind early in processing. Interface stage F is spatiotemporal network sampling: point or grid sampling of the model wind field is performed using the event-conditional observation network layout; Moore's MIST (2.3 km), MICRO (5.2 km), and MESO (33.4 km) synthetic networks and their sampling probabilities (43%, 18%, 0.7%) and minimum resolvable areas (21.2, 108.2, 4462.2 km^2) provide direct evidence for this sampling interface, and Figure 7 shows the percentages at which each of the three network classes successfully samples various wind thresholds (Moore 2024). This step produces event-conditional point or grid evidence, and its detection probability depends strongly on swath area, station spacing, and the severe subset, and cannot be treated as a universal detection guarantee.

Interface stage G is height, averaging-period, sampling, and roughness transforms: the model-level or 10 m diagnostic wind is transformed to the engineering reference height, engineering averaging period, and specified roughness, with each transform explicitly modeled and its uncertainty constrained. The height transform must handle the non-logarithmic near-surface profile of downburst outflow; the log law of the stationary boundary layer cannot be applied. The averaging-period transform must convert the model instantaneous or 60 s quantity to the engineering 3 s averaging period; this conversion has no ready gust factor for nonstationary outflow and must be calibrated against observations. The roughness transform must map the idealized

semislip or specified land-use lower boundary to the actual roughness of the engineering site. The sampling transform must convert the gridpoint envelope to the fixed-point time-series extreme. Moore's self-declared semislip friction bias, terrain neglect, and peak under-resolution all inject uncertainty at this step that must be propagated (Moore 2024). Each transform is an independent error source whose uncertainty must be propagated link by link rather than offset with a single empirical coefficient in one step; otherwise the final engineering gust will conceal its internal multiple unquantified assumptions. Interface stage H is engineering gusts and downstream use: only after the preceding transforms are explicitly modeled and uncertainty is bounded can `Vhat_3s` be discussed, and `Vhat_3s` together with downstream extreme value analysis (EVA) or structural loading use are independent scientific modules requiring PI approval (Qu and Ji 2013). As of the current status, CM1 can provide self-consistent three-dimensional wind fields, near-surface diagnostics, and sampling statistics under event conditions, and can support sampling-probability and footprint discussion in a manner with clear source boundaries; however, the height transform, averaging-period transform, roughness transform, `Vhat_3s` conversion, EVA, and structural loading from model diagnostics to engineering gusts are all missing and unapproved modules that must each be calibrated against observations and have their uncertainties propagated. Three milestones can mark progress along this path. Achieving a Moore-level benchmark requires reproducing Moore's ensemble statistics (severe-wind frequency, 50-to-10 m ratio, duration, coverage, and sampling probability) under a frozen configuration and passing independent CM1 technical review; this establishes methodological and statistical reproducibility, not engineering accuracy. Achieving a PETW validation requires quantitative comparison of CM1 near-surface diagnostics against event-level observations at matched height, averaging period, and spatial scale with stated tolerances; this establishes event-level near-surface quantitative credibility. Achieving engineering readiness requires the above near-surface transforms and `Vhat_3s` conversion to be explicitly modeled with bounded uncertainty and approved, and downstream EVA and structural loading modules to be independently completed; this establishes the final chain usable for design. None of these three currently holds, and they must be established in sequence and cannot be skipped. This section therefore gives an operable path while maintaining `not_approved`: it is usable for research and internal review, providing a structured interface for subsequent experiment matrices and validation, but it cannot be used for design values, structural loading, or publication conclusions.

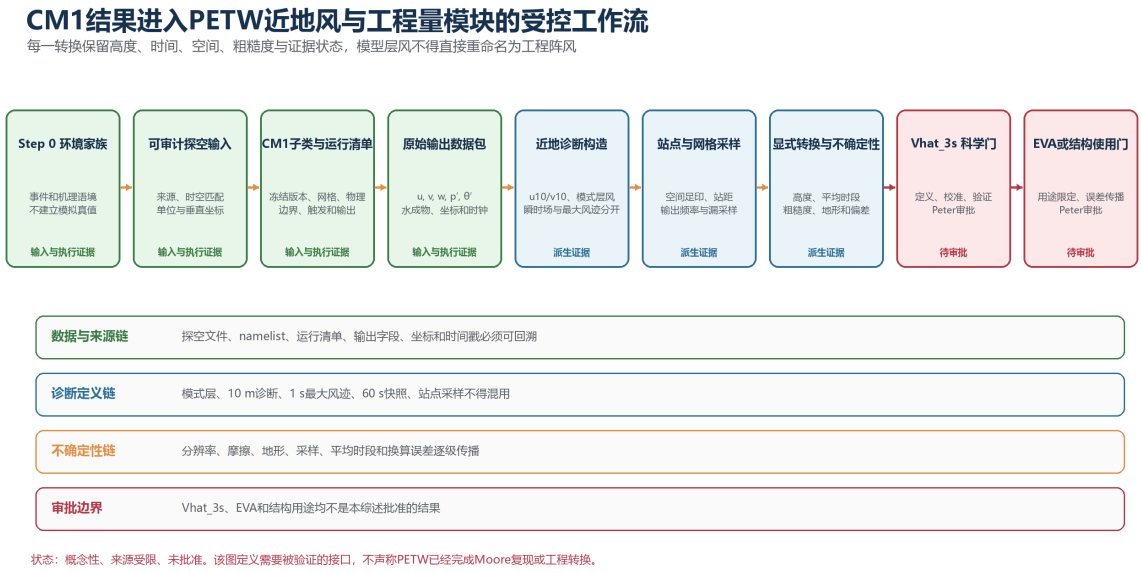


Figure 6: Schematic of the PETW usable but unapproved workflow from event and mechanism scenario through CM1 simulation to engineering gusts. This figure is an original schematic newly synthesized from the literature cited in this study and reproduces no source figure. The concepts it synthesizes are drawn from Moore (2024) near-surface diagnostics and sampling interface (Moore 2024), event-paired and observational constraints (Schumacher et al. 2023; Childs et al. 2021), and engineering hazard background (Qu and Ji 2013). The figure sequentially shows event and mechanism scenario, sounding source and conversion, CM1 subcategory and frozen namelist, raw output, near-surface processing, network sampling, height and averaging-period and roughness transforms (with bounded uncertainty), and `Vhat_3s` and EVA or structural use; every missing transform and unapproved module is explicitly marked in the figure, and the overall status is `not_approved`.

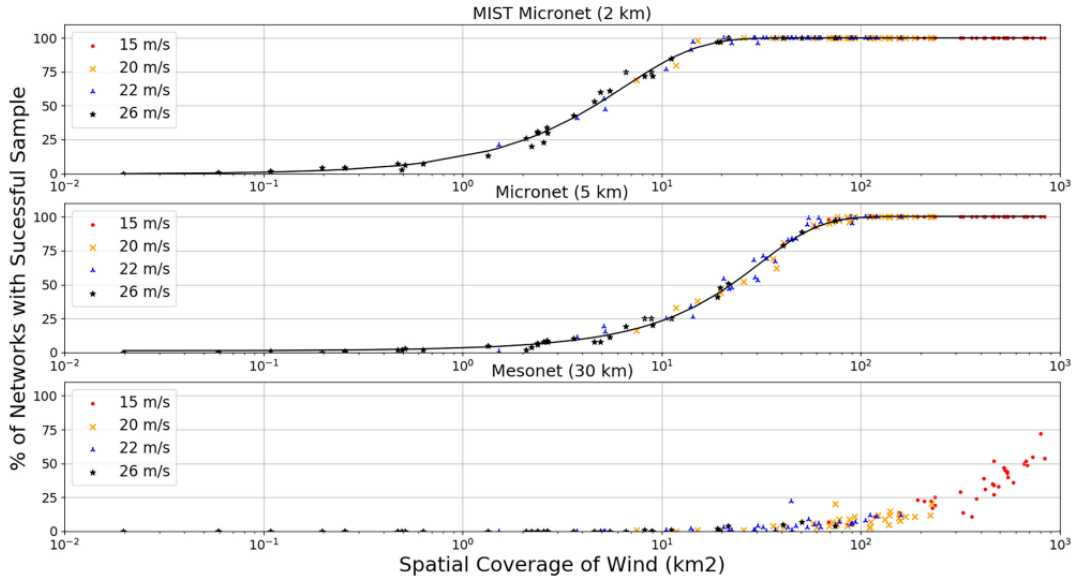


Figure 7: Percentages at which the MIST, MICRO, and MESO simulated observation networks successfully sample various wind thresholds, together with logistic fits for MIST and MICRO, from Moore (2024) (Zotero key JFT4WRDC), original Figure 13, located on page 16 of that paper's PDF. The scientific function is to support the PETW sampling-interface discussion; the results are conditional on idealized swaths, the 50 m diagnostic, station spacings, and the severe subset. This is an internal-review crop with status `internal_review_only_permission_or_redraw_required_before_publication`; copyright permission must be obtained or the figure redrawn before formal publication, and this study does not imply that permission has been obtained.

12 Discussion, Research Gaps, and Roadmap

An overall assessment of CM1 downburst research should focus on its internal tensions rather than recapitulating results. The first tension is between realism and controllability: wet full-storm microphysics simulations trade physical self-consistency for high sensitivity to sounding and trigger, while cooling-source methods trade controllability for sacrificing real microphysics; the two hold opposite ends of the same problem, and their results cannot be migrated between one another. The second tension is between ensemble and event reconstruction: Moore's ensemble characterizes distributions with statistical representativeness but does not correspond to any single real event; the WRF event line reconstructs real events but cannot resolve near-surface peaks. Together they demonstrate that statistical representativeness and event-level accuracy are difficult to achieve simultaneously under current resolution and computational conditions. The third tension is between resolution and scale coverage: grid sensitivity shows that increasing spacing, while only slightly lowering peak wind, produces a stronger downburst and a larger outflow radius, suggesting that peak wind may be under-resolved, while finer grids limit domain size and ensemble scale (Moore 2024; Bryan et al. 2003). The fourth tension is between physical closure and empirical forcing: Deardorff TKE and other subgrid closures together with semislip friction are not uniquely sufficient for downbursts, and their near-surface effects still require dedicated evaluation (Moore 2024; Bryan and Fritsch 2002).

The fifth tension is between the environmental representativeness of the ensemble and the site-specificity of engineering: Moore's ensemble represents the statistics of a particular weakly forced moist convection environment family, while engineering design targets a specific site and a specific return period; the mapping between the two is not trivial, and ensemble statistics can provide a prior distribution but cannot replace event assessment under site conditions. The sixth tension is between computational cost and experimental scale: finer grids, higher output frequencies, and larger domains all increase computational cost, and ensemble scale, parameter-sweep range, and completeness of sensitivity experiments are all constrained by this; any PETW experiment matrix must balance fidelity against feasible scale.

Research gaps concentrate in four directions. Regarding the validation ceiling, event-matched pointwise quantitative validation and unified tolerance do not currently exist; validation remains primarily at the level of morphological plausibility and source-specific metrics, and this is an upper bound that all downstream uses must face (Moore 2024; Hawbecker et al. 2021). Regarding near-surface transfer, the height, averaging-period, roughness transforms and `Vhat_3s` conversion from model diagnostics to engineering gusts are all missing and must each be calibrated against observations (Qu and Ji 2013). Regarding terrain and exposure, idealized simulations universally neglect terrain, and the local enhancement of near-surface wind by terrain cannot be ignored and must be supplemented with sensitivity experiments. Regarding reproducibility, most studies have namelists and run packages that are available upon request or NR, and PETW reproduction must first establish a frozen runnable package containing namelist, source modifications, initial soundings, and a run manifest. These four directions collectively define the distance between current evidence and engineering readiness: the deficit is not in qualitative understanding but

in event-matched quantitative validation, auditable near-surface transforms, quantification of terrain and exposure, and runnable reproduction packages. The minimum experiment matrix for PETW should cover at least: a set of same-sounding grid-sensitivity experiments to constrain the magnitude of peak under-resolution; a set of friction and roughness sensitivity experiments to quantify near-surface wind bias; a set of terrain-sensitivity experiments to assess exposure effects; a set of high-output-frequency experiments to constrain temporal-sampling missed peaks; and a set of comparisons with observations at matched height, averaging period, and spatial scale to establish a usable quantitative tolerance. These experiments collectively constitute the minimum path from current evidence to a Moore-level benchmark and PETW validation; the pass criteria and uncertainty budget for each step must be explicitly recorded, and the engineering conversion chain must remain independent and unapproved pending PI scientific review.

13 Concluding Remarks

This study takes the CM1 model, operating within a cloud-resolving, nonhydrostatic, compressible moist equation set framework, as its central thread and systematically reviews the formation mechanisms and basic dynamics of downbursts, the historical evolution of simulation method families, the detailed methodological framework and methodological subcategories of CM1, research progress centered on the Moore (2024) ninety-seven-member ensemble, the reproducibility of initial and numerical settings, the validation tiers reached by comparisons of simulations with observations and physical experiments, and every mismatch in near-surface wind-field diagnostics. The evidence base shows that CM1 is suited to the downburst problem in its explicit solving of downdraft generation, outflow evolution, and near-surface structure; the Moore ensemble and ten direct CM1 studies provide source-bounded methods, structure, and statistics in the directions of near-surface peaks, peak factors, scaling laws, numerical schemes, event mechanisms, and outflow interaction; however, current validation remains primarily at the level of morphological plausibility and source-specific metrics, event-matched pointwise quantitative validation and unified tolerance do not yet exist, and therefore this study does not provide any single universal accuracy percentage, and all quantitative results retain their configuration, diagnostic definition, sample population, and limitations.

Regarding the PETW interface, this study presents an operable path from event and mechanism scenario through CM1 simulation to near-surface diagnostics, network sampling, and engineering gusts, step by step marking its source-bounded evidence and missing transforms. Between CM1 output and engineering gusts, height transform, averaging-period transform, roughness transform, `Vhat_3s` conversion, and downstream extreme value analysis and structural loading are all independent and unapproved scientific modules that must each be calibrated against observations and have their uncertainties propagated. The interface provided by this study maintains `not_approved` status while remaining operable, and its final status is `pending_peter_scientific_review`; any design values, structural loading, publication conclusions, or main-report integration require subsequent scientific review and approval.

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